Computational Methods in Chemical Engineering

List of Experiments

Sr. No.	Title of Experiment
1	Attainable region for PFR & CSTR for Van De Vusse reaction system
2	Reactor network synthesis for manufacture of maleic anhydride
3	Minimum utility target and pinch point using temperature interval method
4	Minimum utility target and pinch point using HCC & GCC method in MS Excel
5	Minimum utility target and pinch point using LP in MS Excel
6	Sequencing of multiple distillation columns
7	Design and scheduling of batch process
8	Minimum utility target and pinch point and design of heat exchanger network using Software

Experiment No. 1: Attainable region for PFR & CSTR for Van De Vusse reaction system

Aim: To determine the attainable region for PFR and CSTR for the Van De Vusse reaction system.

Theory

For CSTR:

Finding the CSTR effluent concentration C requires the solution of a system of nonlinear equations.

Design Equation: $\boldsymbol{c_i = c_{if} + \tau r_i(C)}$

Matrix form: $\begin{bmatrix} c_A \\ c_B \\ . \\ . \\ . \\ c_n \end{bmatrix} = \begin{bmatrix} c_{Af} \\ c_B f \\ . \\ . \\ . \\ c_{nf} \end{bmatrix} + \tau \begin{bmatrix} r_A(C) \\ r_B(C) \\ . \\ . \\ . \\ r_n(C) \end{bmatrix}$

Vector form: $C = C_f + \tau r(C)$ gives $\left(C - C_f\right) = \tau r(C)$

Rate vector evaluated at a CSTR effluent concentration C, operated with a CSTR feed concentration of C_f is colinear to the CSTR mixing vector $(C - C_f)$.

For PFR:

System of ordinary differential equations (ODEs), solution an initial condition (feed concentration) and integration time (PFR residence time).
Solving the system of ODEs results in a PFR solution trajectory.

Design equation: $\boldsymbol{\frac{dc_i}{d\tau} = r_i(C)}$

Matrix form: $\begin{bmatrix} \frac{dc_A}{d\tau} \\ \frac{dc_B}{d\tau} \\ . \\ . \\ . \\ \frac{dc_n}{d\tau} \end{bmatrix} = \begin{bmatrix} r_A(C) \\ r_B(C) \\ . \\ . \\ . \\ r_n(C) \end{bmatrix}$

Vector form: $\frac{dC}{d\tau} = r(C)$

Mixing vector $(C - C_f)$ is a scalar multiple of vector r(C). Gradient of PFR trajectory with respect to PFR residence time evaluated at C.

Reference: https://attainableregions.com/examples/ar_fundamentals.html

Problem Statement

Determine the attainable region for the van de Vusse (1964) reaction system, consisting of the reactions:

$$\boldsymbol{A} \underset{k_2}{\overset{k_1}{\Leftrightarrow}} \boldsymbol{B} \overset{k_3}{\rightarrow} \boldsymbol{C}$$

$$\boldsymbol{2A} \overset{k_4}{\rightarrow} \boldsymbol{D}$$

Where rate constants are,
$k_1 = 0.01\ s^{-1}$
$k_2 = 5\ s^{-1}$
$k_3 = 10\ s^{-1}$
$k_4 = 100\ m^3/kmol.\ s^{-1}$

Reactions 1, 2, and 3 are first-order in A, B, and B, respectively, while reaction 4 is second-order in A.

Procedure:

From Warren D. Seider, J. D. Seader, Daniel R. Lewin, Product and Process Design Principles: Synthesis, Analysis, and Evaluation

Step 1:

- First construct a trajectory for a PFR from feed point to complete conversion of A or chemical equilibrium by solving simultaneously the kinetic equations for A and B, where τ is the PFR residence time.
- The trajectory in C_A - C_B space is plotted as curve ABC for case that Component A is completely converted.

$$\frac{dC_A}{d\tau} = -k_1 C_A + k_2 C_B - k_4 C_A^2$$

$$\frac{dC_B}{d\tau} = -k_1 C_A + k_2 C_B - k_3 C_B$$

Step 2:

- When the PFR trajectory bounds a convex region, this constitutes a candidate attainable region.
- A convex region is one in which all straight lines drawn from one point on the boundary to any other point on the boundary lie wholly within the region or on the boundary.
- When the rate vectors at concentrations outside of the candidate AR do not point back into it, the current limits are the boundary of the AR and the procedure terminates.
Rate vectors: $[dC_A/d\tau\, , dC_B/d\tau]^T$
- Here the PFR trajectory is not convex from A to B, so proceed to the next step to determine if the attainable region can be extended beyond the curve ABC.

Step 3:

- The PFR trajectory is expanded by linear arcs, representing mixing between the PFR effluent and the feed stream, extending the candidate attainable region.
- A linear arc, ADB, is added, extending the attainable region to ADBC. Rate vectors computed along this arc are found to point out of the extended AR, proceed to the next step.

Step 4:

- After placing CSTR trajectory that extends AR most, additional linear arcs that represent mixing of streams are placed to ensure that AR remains convex.
- The CSTR trajectory is computed by solving the CSTR form of the kinetic equations for A and B.
- The CSTR trajectory that extends the AR most is that computed from feed point, at C_{A0}, largest concentration of A: curve AEF which passes through point B.
- Since the union of the previous AR and the CSTR trajectory is not convex, a linear arc, AGO, is augmented.
- This arc represents a CSTR with a bypass stream.

Step 5:

- A PFR trajectory is drawn from the position where the mixing line meets the CSTR trajectory.
- If this PFR trajectory is convex, it extends the previous AR to form an expanded candidate AR. Then return to Step 2. Otherwise, repeat the procedure from Step 3.
- The PFR trajectory, OHI leads to a convex attainable region.
- The boundaries of region are: (a) the linear arc, AGO (CSTR with bypass stream) (b) the point O (CSTR) and (c) the arc OHI (CSTR followed by a PFR in series).
- The maximum composition of B is obtained at point H, using a CSTR followed by a PFR.

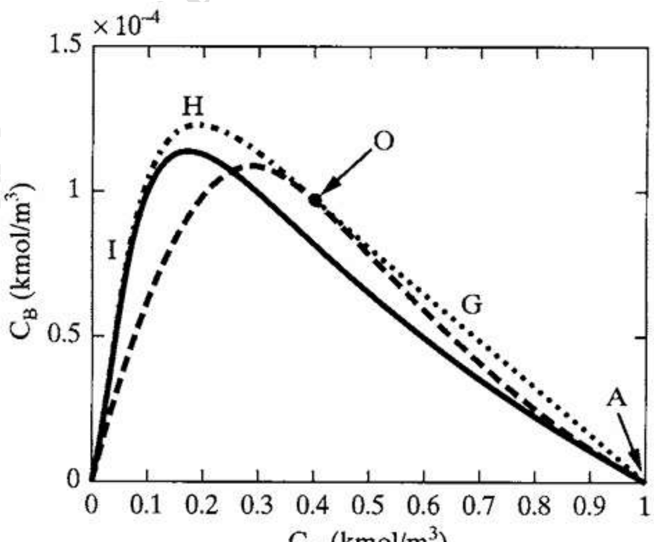

Figure 1: Construction of attainable region for the van der Vusse reaction: addition of PFR in series with CSTR (dot-dashed line).

Solution:

- The boundary of the attainable region is composed of arcs, each of which results from the application of a distinct reactor type.
- With a feed of 1 kmol/m^3 of A, AR boundary is composed of an arc representing a CSTR with bypass (curve C), a CSTR (point 0), and a CSTR followed by a PFR (curve D).
- Within the region bounded by the three arcs and the horizontal base line ($C_B = 0$), product compositions can be achieved with some combination of these reactor configurations.

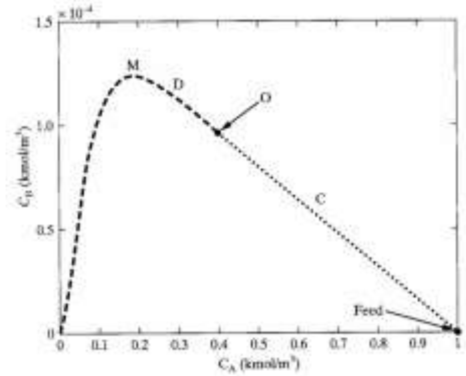

Figure 2. Attainable region for van de Vusse reaction.

- The appropriate reactor configuration along the boundary of the attainable region depends on the desired effluent concentration of A.
- 1 > C_A > 0.38 kmol/m^3, a CSTR with bypass (curve C) provides the maximum concentration of B,
- C_A < 0.38 kmol/m^3 max. X using a CSTR (point 0), followed by a PFR (Curve D).
- Maximum achievable concentration, $C_B = 1.25 \times 10^{-4}$ kmol/m^3, is obtained using a CSTR followed by a PFR (at point M along curve D).

Simulation:
MATLAB code for First order parallel reaction in PFR

Reference: Kompally vishal kumar (2025). First order parallel reaction in PFR (https://www.mathworks.com/matlabcentral/fileexchange/92965-first-order-parallel-reaction-in-pfr), MATLAB Central File Exchange. Retrieved June 3, 2025.

Functions:

- parallel_reaction
- parallel_reaction_sol.m

Code: parallel_reaction_sol.m

```
function y=parallel_reaction_sol(k1,k2,ca0,tspan)
ca0=1; %mol/m3
c0=[ca0;0;0];
tspan = 5;
tspan=[0 tspan];
k1=3;k2=2;
```

```
[tsol,ysol]=ode45(@(t,c) parallel_reaction_ode(t,c,k1,k2),tspan,c0);
y.time=tsol;
y.concentration=ysol;

figure(1)
plot(tsol,ysol)
legend('Ca','Cb','Cc')
xlabel('Time (seconds)')
ylabel('Concentration (mol/m^3)')

end

function y=parallel_reaction_ode(~,c,k1,k2)
y=zeros(3,1);
y(1)=-k1*c(1);
y(2)=k1*c(1)-k2*c(2);
y(3)=k2*c(2);
end
```

Link: https://matlab.mathworks.com/open/fileexchange/v1?id=92965

Result:

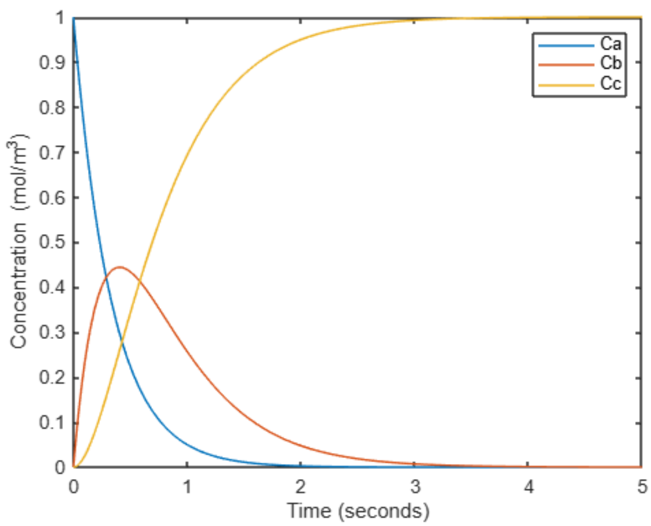

Figure 3. Concentration profiles of A, B, C against time for PFR.

Experiment:

1. Vary the initial concentration of A and observe its effect on the concentration profiles.
2. Vary the reaction time and observe its effect on the concentration profiles.
3. Vary the reaction rates k_1, k_2 and observe its effect on the concentration profiles.
4. Make a report of these results and attach the printout in the manual.

MATLAB code for First order parallel reaction in CSTR

Reference: https://in.mathworks.com/matlabcentral/answers/702322-cstr-reaction-model-yield vs-conversion-graph

Code:

```
clear all; close all
clc
%CONTINUOUS STIRRED TANK REACTOR CODE CONCENTRATION AS A FUNCTION OF CSTR
%TIME
%time span specification
timeStart = 0; timeEnd   = 600;%s
timeSpan  = [timeStart:0.1:timeEnd];
cao = 0.13;
%initial condition specification
initCA = 0.13; initCB = 0.1; initCC =  0 %mol/m^3
initCon= [initCA, initCB, initCC];
v0A = 2.5*10^-5;
%v0A = 1.93*10^-5; %L/s
k2  = 0.86;
%ODE solver CSTR
[time, Ccstr] = ode45(@diffcstr11, timeSpan, initCon,[], v0A, k2);
%plot CSTR
figure (1)
plot(time, Ccstr, 'Linewidth',1.5)
grid on
xlabel('time [s]')
ylabel('concentration [mol/m^3]')
legend('c_A','c_B','c_C')

function  dcdt = diffcstr11(t, C, v0A, k2)
% C(1) is cA, C(2) is cB, C(3) is cC
% parmeters
cA0  = 0.13;   %mol/L
cB0  = 0.1;   %mol/L
V    = 30*10^-3; %L
k1   = 1; %L/(mol*s)
k2   = 0.15; %L/(mol*s)
%v0A = 0.03; %L/s
v0B = (5*10^-5); %L/s
v0  = v0A+ v0B;%L/s
tau = V/v0;
%Rate equations
rA = -(k1*C(1)*C(2))-(k2*C(1)*C(2));
rB = rA;
rC = (k1*C(1)*C(2));
% differential equations
dcdt = [ v0A*(cA0/V) - (C(1)/tau) - (-rA);
         v0B*(cB0/V) - (C(2)/tau) - (-rB);
          -(C(3)/tau) + rC;
```

];

end

Result:

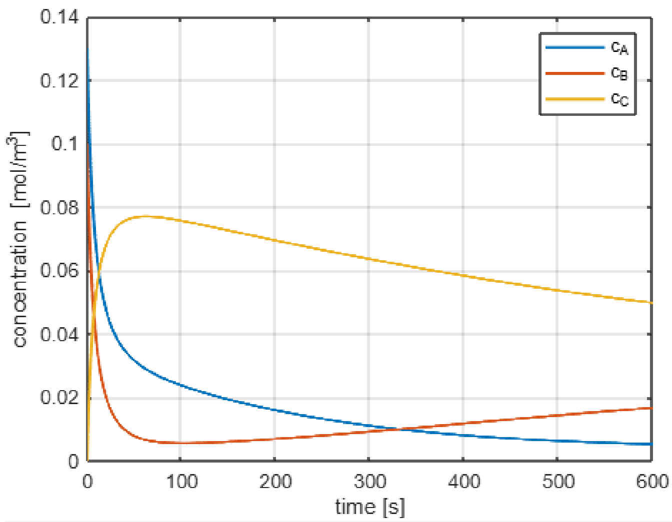

Figure 4. Concentration profiles of A, B, C against time for CSTR.

Experiment:

1. Vary the initial concentration of A and B and observe its effect on the concentration profiles.
2. Vary the reaction time and observe its effect on the concentration profiles.
3. Vary the reaction rates k_1, k_2 and observe its effect on the concentration profiles.
4. Make a report of these results and attach the printout in the manual.

Using MATLAB software to generate and plot data to obtain attainable region for the Van De Vusse reaction system.

Reference code: Housam Binous (2025). Construction of Attainable Region for van de Vusse Kinetics (https://www.mathworks.com/matlabcentral/fileexchange/8878-construction-of-attainable-region-for-van-de-vusse-kinetics), MATLAB Central File Exchange. Retrieved June 3, 2025.

Basis: Feed consists of 1 kmol/m^3 of A

Functions: **Van de Vusse/**

1. Main_AR_VandeVusse.m
2. cstr(x)
3. vandevusse(t,x)

Code: Main_AR_VandeVusse.m

```
% Construction of Attainable Region for van de Vusse Kinetics
% Author's Data: Housam BINOUS
% Department of Chemical Engineering
% National Institute of Applied Sciences and Technology
% Tunis, TUNISIA
% Email: binoushousam@yahoo.com
global tau
x0=[1 0];
[t,x]=ode45(@vandevusse,[0 10],x0);
Grad1=diff(x(:,2))./diff(x(:,1));
for i=1:1000
   r(i)=0.001*i;
   f(i)=interp1(x(1:220,1),Grad1(:,1),r(i))*(r(i)-1)-interp1(x(:,1),x(:,2),r(i));
end
f=f';
r=r';
z=interp1(f(100:980),r(100:980),-3e-7);
figure(1)
plot(x(:,1),x(:,2),'r')
axis([0 1 0 15e-5])
line([1 z],[0 interp1(x(:,1),x(:,2),z)])
j=1;
figure(2)
for i=1:10:10000
   tau=0.0001*i;
   options=optimset('display','off');
   X=fsolve(@cstr,[0.2 0.4],options);
   C(j,1)=X(1);
   C(j,2)=X(2);
   j=j+1;
end
j=1;
for i=1:100
   tau=i;
   options=optimset('display','off');
   X=fsolve(@cstr,[0.2 0.4],options);
   C2(j,1)=X(1);
   C2(j,2)=X(2);
   j=j+1;
end
Grad=diff(C(:,2))./diff(C(:,1));
for i=1:1000
   r(i)=0.001*i;
   f(i)=interp1(C(1:999,1),Grad(:,1),r(i))*(r(i)-1)-interp1(C(:,1),C(:,2),r(i));
end
z=interp1(f(100:980),r(100:980),-0.0004e-3);
plot(x(:,1),x(:,2),'r')
axis([0 1 0 15e-5])
hold on
```

```
plot(C(:,1),C(:,2),'g')
line([1 z],[0 interp1(C(:,1),C(:,2),z)])
plot(C2(:,1),C2(:,2),'g')
x0=[z interp1(C(:,1),C(:,2),z)];
[t,x]=ode45(@vandevusse,[0 10],x0);
plot(x(:,1),x(:,2),'m')
```

Link: https://matlab.mathworks.com/open/fileexchange/v1?id=8878

Code: cstr(x)

```
% Construction of Attainable Region for van de Vusse Kinetics
% Author's Data: Housam BINOUS
% Department of Chemical Engineering
% National Institute of Applied Sciences and Technology
% Tunis, TUNISIA
% Email: binoushousam@yahoo.com
function f=cstr(x)
global tau
k1=0.01;k2=5;k3=10;k4=100;
f(1)=1-x(1)-tau*(k1*x(1)-k2*x(2)+k4*x(1)^2);
f(2)=x(2)-tau*(k1*x(1)-k2*x(2)-k3*x(2));
```

Code: vandevusse(t,x)

```
% Construction of Attainable Region for van de Vusse Kinetics
% Author's Data: Housam BINOUS
% Department of Chemical Engineering
% National Institute of Applied Sciences and Technology
% Tunis, TUNISIA
% Email: binoushousam@yahoo.com

function xdot=vandevusse(t,x)

k1=0.01;k2=5;k3=10;k4=100;

xdot(1)=-k1*x(1)+k2*x(2)-k4*x(1)^2;
xdot(2)=k1*x(1)-k2*x(2)-k3*x(2);

xdot=xdot';
```

Result:

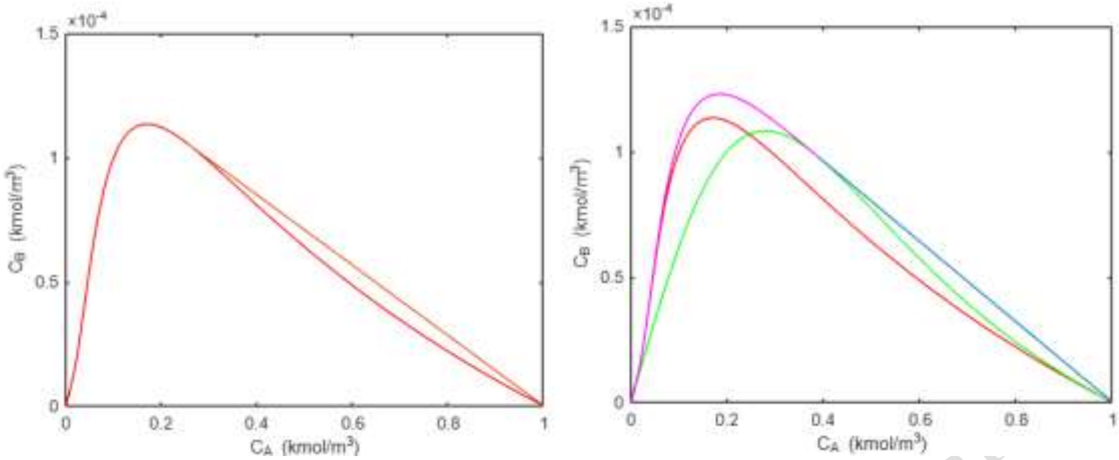

Figure 5. Construction of the attainable region for the van de Vusse reaction: PFR trajectory with mixing line, CSTR trajectory, addition of bypass to CSTR, addition of PFR in series with CSTR.

Experiment:

Using the codes for PFR and CSTR, add the van De Vusse reactions in the codes, find the concentration profiles and plot C_A vs. C_B.

Make a report of these results and attach the printout in the manual. Submit the MATLAB codes by email before/on submission date.

Conclusion:

Experiment No. 2: Reactor network synthesis for manufacture of maleic anhydride

Aim: To synthesis reactor network for manufacture of maleic anhydride.

Problem Statement: Example 6.4 from Textbook

Maleic anhydride, $C_4H_2O_3$, is manufactured by the oxidation of benzene with excess air over Vanadium Pentoxide catalyst (Westerlink and Westerterp, 1988). The following reactions occur:

Reaction I: $C_6H_6 + 9/2\ O_2 \rightarrow C_4H_2O_3 + 2CO_2 + 2H_2O$

Reaction 2: $C_4H_2O_3 + 3O_2 \rightarrow 4CO_2 + H_2O$

Reaction 3: $C_6H_6 + 15/2\ O_2 \rightarrow 6C0_2 + 3H_2O$

Since air is supplied in excess, the reaction kinetics are approximated using first-order rate laws:

$$\begin{array}{l} A \xrightarrow{r_1} P \xrightarrow{r_2} B \\ A \xrightarrow{r_3} C \end{array}$$

$r_1 = k_1CA$, $r_2 = k_2C_P$, $r_3 = k_3C_A$

In the above, A is benzene, P is maleic anhydride (the desired product), and B and C are the undesired byproducts (H_2O and CO_2). The rate coefficients are (in m^3/kg catalyst·s)

$k_1 = 4{,}280 \exp [-12{,}660/T(K)]$

$k_2 = 70{,}100 \exp [-15{,}000/T(K)]$

$k_3 = 26 \exp [-10{,}800/T(K)]$

Given that the available feed stream contains benzene at a concentration of 10 mol/m^3 with a volumetric flow rate, v_0, of 0.0025 m^3/s (the feed is largely air), determine a network of isothermal reactors to maximize the yield of maleic anhydride.

Procedure:

Example 6.4 From Warren D. Seider, J. D. Seader, Daniel R. Lewin, Product and Process Design Principles: Synthesis, Analysis, and Evaluation

- First, an appropriate reaction temperature is selected. Following Heuristic 7 from Chapter 5, Figure 6 shows the effect of temperature on the three rate coefficients, and indicates that in the range $366 < T < 850$ K, the rate coefficient, k_1 of the desired reaction to MA is larger than those of the competing reactions. An operating temperature at the upper end of this range is recommended, as the rate of reaction increases with temperature.

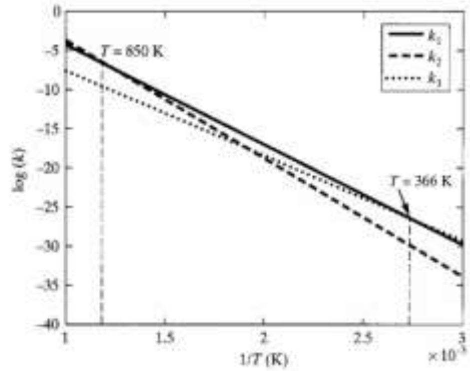

Figure 6. Effect of temperature on rate constants of MA reaction.

- Since all of the reaction rate expressions involve only benzene and maleic anhydride, the system can be expressed in a two-dimensional composition space. For this system, the attainable region is straight forward to construct.
- This begins by tracing the composition space trajectory for a packed bed reactor (PBR), modeled as a PFR, which depends on the solution of the molar balances, where W is the kg of catalyst.
- In the above equations, the temperature-dependent rate constants are computed using Eqs. Given in problem statement.

$$\boldsymbol{v_0 \frac{dC_A}{dW} = -k_1 C_A - k_3 C_A \quad , C_A(0) = C_{A0}} = 10 \text{ mol/m}^3$$

$$\boldsymbol{v_0 \frac{dC_P}{dW} = -k_1 C_A - k_2 C_P \quad , C_P(0)} = 0 \text{ mol/m}^3$$

- Figure 7 presents solutions of these equations as trajectories in C_A-C_P space for several operating temperatures. Since these trajectories are convex, and rate vectors computed along their boundaries are tangent to them, it is concluded that each trajectory bounds the attainable region for its corresponding temperature.
- Evidently, a single plug-flow reactor (or packed-bed reactor in this case) provides the maximum production of maleic anhydride, with the required space velocity being that which brings the value of C_P to its maximum.
- At 800 K, it is determined that the maximum concentration of maleic anhydride is 3.8 mol/m^3, requiring 4.5 kg of catalyst. At 600 K it is 5.3 mol/m^3, but at this low temperature, 1,400 kg of catalyst is needed. A good compromise is to operate the PBR at an intermediate temperature, for example, 770 K, with a maximum concentration of maleic anhydride of 4.0 mol/m^3, requiring 8 kg of catalyst.
- Figure 8 (a) shows composition profiles for all species as a function of bed length (proportional to the weight of catalyst), indicating that the optimal catalyst loading, where the concentration of maleic anhydride is a maximum, is about 8 kg for isothermal operation at 770 K.
- Figure 8 (b) indicates that the yield (the ratio of the desired product rate and feed rate) under these conditions is 61%, while the selectivity is only about 10%. The selectivity (the ratio of the desired product and total products) for this reaction system is poor, due to the large amounts of CO_2 and H_2O produced, with the highest selectivity, achieved by repressing both of the undesired reactions, at 22%.

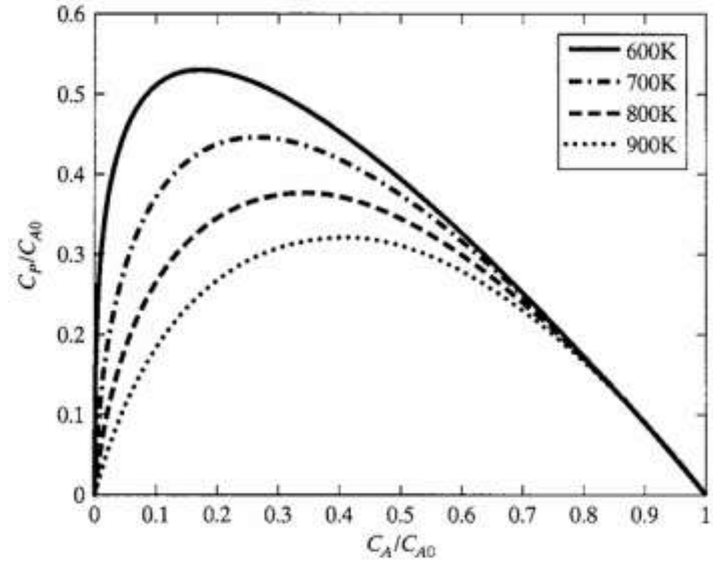

Figure 7. Attainable regions for MA production at various temperatures.

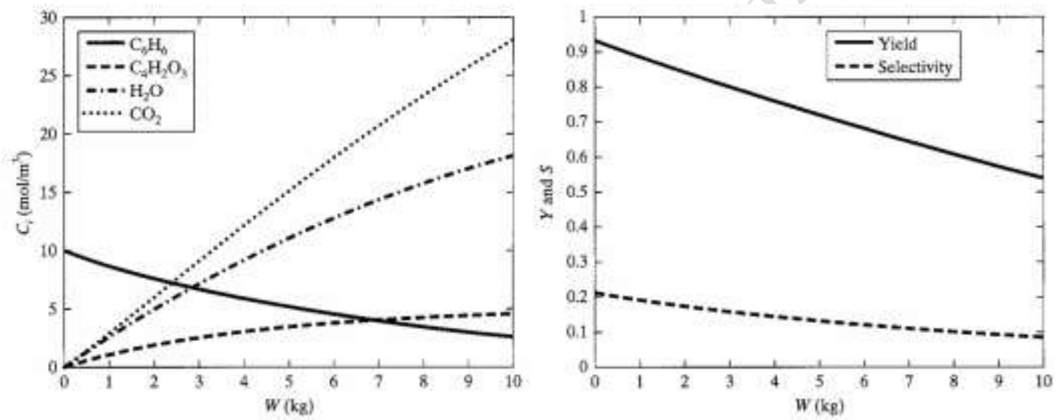

Figure 8. (a) Composition profile and (b) Selectivity and Yield for MA production at 770 K.

Simulation Experiment:
Using MATLAB software and codes processed in experiment 1.

- Using the rate constant equation for dependance on temperature and plot log(k) vs. 1/T for a range of temperatures from 100-1000 K. Observe if plot obtained is similar to that of the textbook.

  k_1 = 4,280 exp [-12,660/T(K)]

  k_2 = 70,100 exp [-15,000/T(K)]

  k_3 = 26 exp [-10,800/T(K)]

- Take the temperature range of 600-900 K.
- At one temperature, modify the PFR code and the van de Vusse code and add the PBR design equations modeled as a PFR.

$$\boldsymbol{v_0 \frac{dC_A}{dW} = -k_1 C_A - k_3 C_A} \quad , \boldsymbol{C_A(0) = C_{A0}} = 10 \text{ mol/m}^3$$

$$\boldsymbol{v_0}\frac{\boldsymbol{dC_P}}{\boldsymbol{dW}} = \boldsymbol{-k_1C_A - k_2C_P} \qquad ,\boldsymbol{C_P(0)} = 0 \text{ mol/m}^3$$

- Instead of τ and $dC_A/d\tau$ in the program, replace by W, range of 1 to 10 kg.
- Plot the concentration vs. the weight of catalyst.
- Obtain the attainable region for temperatures of 600, 700, 800, and 900 K.
- Observe what is the optimum temperature and weight of catalyst.
- Make a report of these results and attach the printout in the manual.
- Submit the MATLAB codes by email before/on submission date.

Conclusion:

Experiment No. 3: Minimum utility target and pinch point using temperature interval method

Aim: To determine the minimum utility target and pinch point using the temperature interval method.

Theory

A principal objective in the synthesis of HENs is the efficient utilization of energy in the hot process streams to heat cold process streams. Thus, it is desirable to compute the maximum energy recovery (MER) before synthesizing the HEN and to determine the minimum hot and cold utilities in the network, given the heating and cooling requirements of the process streams. This important first step is referred to as MER targeting, and is useful in that it determines the utility requirements for the most thermodynamically efficient network.

The temperature-interval method was developed by Linnhoff and Flower (1978a, b) follow-ing the pioneering work of Hohmann (1971).

The method is applied to the hot and cold streams, and a systematic procedure unfolds for determining the minimum utility requirements over all possible HENs, given just the heating and cooling requirements for the process streams and the minimum approach temperature in the heat exchangers, ΔT_{min}.

Problem statement (Example 10.2 from textbook):

To introduce this targeting step, the example provided by Linnhoff and Turner (1981) is analyzed, however it involves only sensible heat.

Two cold streams, Cl and C2, are to be heated and two hot streams, Hl and H2, are to be cooled without phase change.

Table 1. Conditions and properties of cold streams C1 and C2 and two hot streams H1 and H2.

Stream	T^s (° F) Source temp.	T^t (° F) Target temp.	mC_p (Btu/(hr.° F))	Q (10^4 Btu/hr)
C1	120	235	20000	230
C2	180	240	40000	240
H1	260	160	30000	300
H2	250	130	15000	180

It is assumed that the heat-capacity flow rate, $mC_p = C$, which is the product of the specific heat and the mass flow rate, does not vary with temperature.

Design a HEN that uses the smallest amounts of heating and cooling utilities possible, such that the closest approach temperature differences never fall below a minimum value. For the temperature range in this example, a reasonable assumption is $\Delta T_{min} = 10°F$. Use the temperature interval method.

Solution (as per textbook):

1. Adjust the source and target temperatures using ΔT_{min}. This is accomplished arbitrarily by reducing the temperatures of the hot streams by ΔT_{min} while leaving the temperatures of the cold streams untouched.
   The adjustment of the hot stream temperatures by subtracting ΔT_{min} brings both the hot and cold streams to a common frame of reference so that when performing an energy balance involving hot and cold streams at the same temperature level, the calculation accounts for heat transfer with at least a driving force of ΔT_{min}.

Table 2. Temperature data of cold and hot streams after adjustment of temperatures by ΔT_{min}.

Stream	**T^s (° F)**	**T^t (° F)**	**Adjusted temperatures**		
C1	120	235	120	235	T_5 T_2
C2	180	240	180	240	T_3 T_1
H1	260	160	250	150	T_0 T_4
H2	250	130	240	120	T_1 T_5

2. The adjusted temperatures are rank ordered, beginning with T_0, the highest temperature. These are used to create a cascade of temperature intervals within which energy balances are carried out.

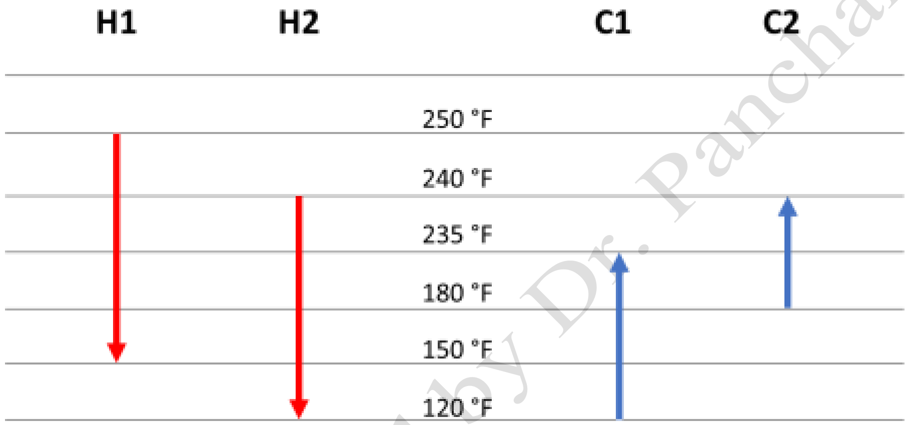

Figure 9. Cascade of temperature intervals with flow of hot and cold stream shown.

For each interval, i, the enthalpy difference, ΔH_i between the energy to be removed from the hot streams and the energy to be taken up by the cold streams in that interval.

$$\Delta H_i = \left(\sum C_h - \sum C_c\right)_i (T_0 - T_i)$$

Table 2. Enthalpy differences for temperature intervals.

Interval, i	**T_{i-1}-T_i (°F)**	$\sum C_h - \sum C_c$ **10^4 Btu/hr.° F**	**ΔH_i 10^4 Btu/hr**
1	250 – 240 = 10	3	30
2	240 – 235 = 5	3 + 1.5 – 4 = 0.5	2.5
3	235 – 180 = 55	3 + 1.5 – 4 – 2 = –1.5	–82.5
4	180 – 150 = 30	3 + 1.5 – 2 = 2.5	75
5	150 – 120 = 30	1.5 – 2 = –0.5	–15

3. Calculate the energy flows between the intervals, assume Q_{steam} = 0 Btu/hr at the start. From the 1st interval add each interval enthalpy to obtain the residual of each interval, as shown below in Figure 10.
4. Find the largest negative residual in the energy flow chart. This value will be the hot utility duty (Q_{steam}). In the final pass, add the Q_{steam} value from the first interval onwards.
5. The interval at which the residual energy is 0 is the pinch point. The final residual energy

obtained is the cold utility duty (Q_c). These obtained values of Q_{steam} and Q_c are the MER targets. The cold pinch temperature is the temperature of the interval (180 °F). The hot pinch temperature is the temperature of interval + ΔT_{min} (190 °F). Draw the diagram for pinch decomposition (Figure 11).

Furthermore, at minimum utilities, no energy flows between intervals, this is referred to as the pinch. To maintain minimum utilities, it is recognized that no energy is permitted to flow across the pinch. To maintain minimum utilities, two separate HENs must be designed, one on the hot side and one on the cold side of the pinch. Energy is added from hot utilities on the hot side of the pinch (50×10^4 Btu/hr), and energy is removed using cold utilities on the cold side of the pinch (60×10^4 Btu/hr), and no energy is permitted to flow across the pinch.

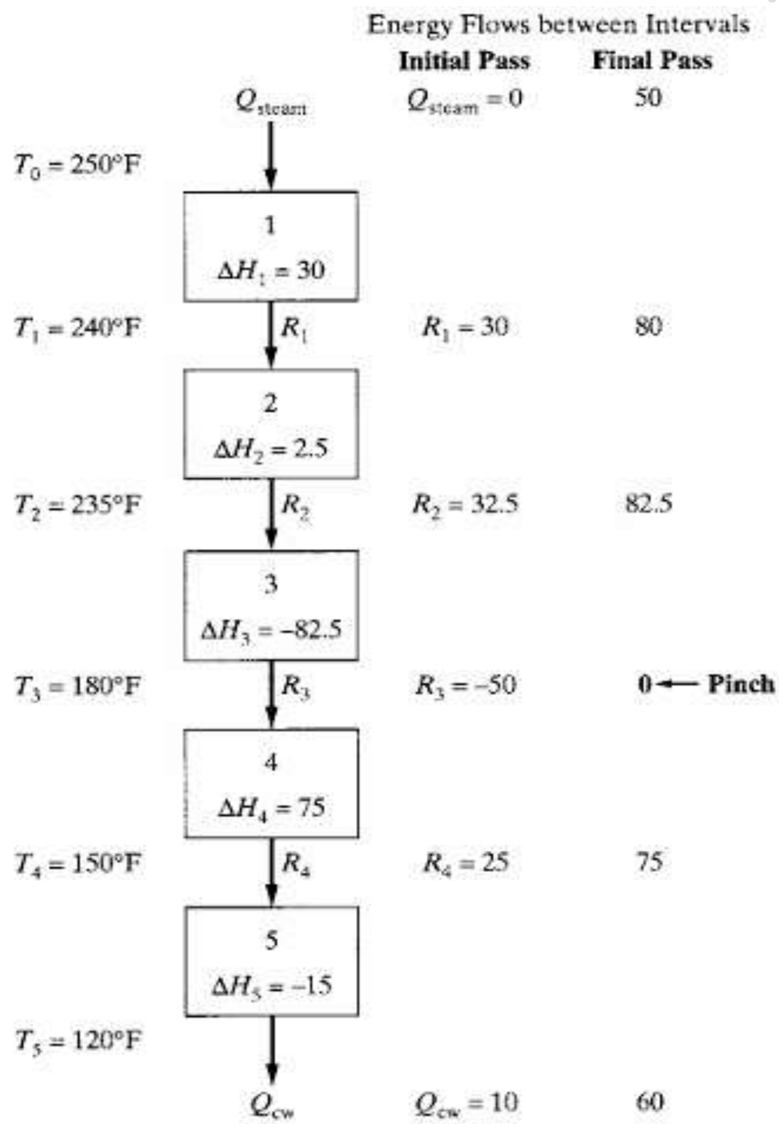

Figure 10. Cascade of temperature intervals, energy balances, and residuals; multiply Q values by 10^4 Btu/hr.

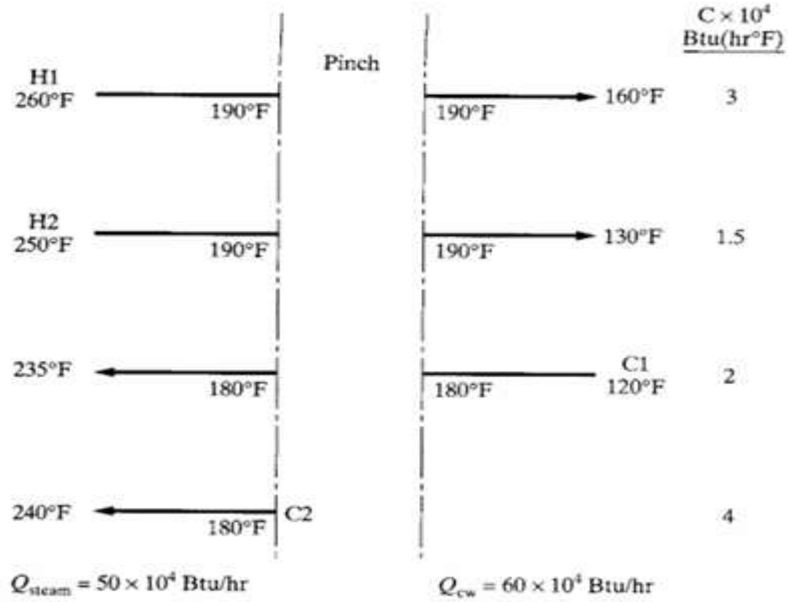

Figure 11. Pinch decomposition of the hot and cold streams.

Table 3. Interval Heat loads (multiply Q values by 10^4 Btu/hr).

	Cooling			Heating		
Interval i	**Temperature Range (° F)**	**Q**	**Cum. Q**	**Temperature Range (° F)**	**Q**	**Cum. Q**
1	250-260	30	480	240-250	0	-
2	245-250	22.5	450	235-240	20	470
3	190-245	247.5	427.5	180-235	330	450
4	160-190	135	180	150-180	60	120
5	130-160	45	45	120-150	60	60

Assignment:

Two cold streams, Cl and C2, are to be heated and two hot streams, Hl and H2, are to be cooled without phase change.

Table 4. Conditions and properties of cold streams C1 and C2 and two hot streams H1 and H2.

Stream	T^s (° C)	T^t (° C)	mC_p (kW)	Q (kW)
C1	40	150	3	?
C2	70	130	1	?
H1	170	40	4	?
H2	200	60	6	?

It is assumed that the heat-capacity flow rate, $mC_p = C$, which is the product of the specific heat and the mass flow rate, does not vary with temperature.

Design a HEN that uses the smallest amounts of heating and cooling utilities possible, such that the closest approach temperature differences never fall below a minimum value.

For the temperature range in this example, a reasonable assumption is $\Delta T_{min} = 10°F$.

Results and discussion:

1. Find the total heating duty and total cooling duty, and their difference.
2. Find the temperature data of cold and hot streams after adjustment of temperatures by ΔT_{min}.
3. Draw the cascade of temperature intervals with flow of hot and cold stream shown.
4. Calculate the Enthalpy differences for temperature intervals.
5. Draw the cascade of temperature intervals, energy balances, and residuals. Find the REMs (minimum energy requirement).
6. Draw the pinch decomposition of the hot and cold streams.

Do these calculations and drawings by hand and submit a report along with manual.

YouTube Link:
https://www.youtube.com/watch?v=7PpysQMD0WE

Conclusion:

Experiment No. 4: Minimum utility target and pinch point using HCC & GCC method in MS Excel

Aim: To determine the Minimum utility target and pinch point using HCC & GCC method in MS Excel.

Theory 1: Hohmann/Lochart Composite Curves (HCC):

The terminology, pinch, is understood more clearly in connection with a graphical display, introduced by Umeda et al. (1978), in which composite heating and cooling curves are positioned no closer than ΔT_{min}.

Composite curves are T/H diagrams (temperature/heat diagrams) used to visualize cold and hot streams and potential heat transfer between them. It is a graphical technique used for visualizing the heat cascade.

$$\boldsymbol{Q = \int_{T_s}^{T_t} C_p dT = C_p(T_t - T_s) = \Delta H}$$

$$\boldsymbol{\frac{dT}{dQ} = \frac{1}{C_p}}$$

The slope of a composite curve is the inverse of the heat-capacity flow rate, C, as C is constant and not a function of T, the curves are straight lines.

For the hot streams, they are cooling curves that begin at the highest temperature and finish at the lowest temperature after the energy has been removed. For the cold streams, they are heating curves that begin at the lowest temperature and finish at the highest temperature after heat has been added.

The hot composite curve has a segment from stream Hl between 260 and 250°F. From 250 to 160°F, both streams Hl and H2 coexist, and hence, their cooling requirements are combined. Note that the combined heat-capacity flow rate is increased, and consequently, the slope of the hot composite curve is reduced. Finally, from 160 to 130°F, only stream H2 appears.

From 120 to 180°F, only stream Cl appears in the cold composite curve. From 180°F to 235°F, the streams Cl and C2 coexist and their heating curves are combined. Finally, from 235 to 240°F, only stream C2 exists.

The composite curves have a closest point of approach of ΔT_{min} = 10°F at the point where stream C2 begins along the cold composite curve; 180°F. The corresponding temperature on the hot composite curve is 190°F. Consequently, these two points provide the temperatures at the pinch.

If ΔT_{min} is reduced to zero, the cold composite curve is shifted to the left until it touches the hot composite curve, this corresponds to an infinite area for heat exchange.

The Hohmann/Lochart Composite Curves (HCC) provides the hot duty Q_{steam}, cold duty Q_{cw} as shown in Figure 12.

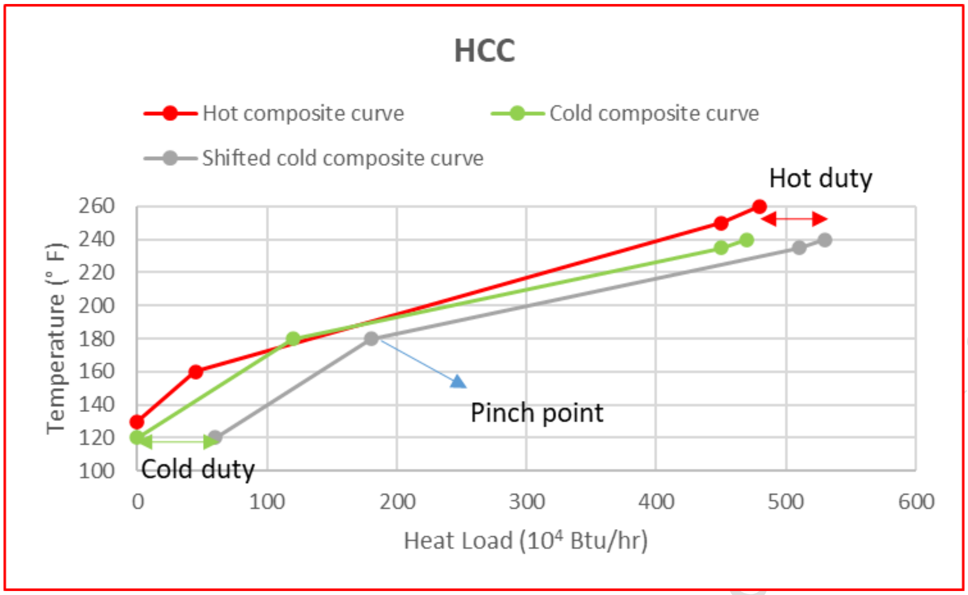

Figure 12. Hohmann/Lochart Composite Curves (HCC) to obtain hot duty Q_{steam}, cold duty Q_{cw} and pinch point.

Problem Statement: (Example 10.3 from textbook)

In this example, the minimum utility requirements for a HEN involving the four streams in Example 10.2 are determined using the graphical approach by Umeda et al. (1978).

Two cold streams, Cl and C2, are to be heated and two hot streams, Hl and H2, are to be cooled without phase change.

Table 5. Conditions and properties of cold streams C1 and C2 and two hot streams H1 and H2.

Stream	T^s (° F) Source temp.	T^t (° F) Target temp.	mC_p (Btu/(hr.° F))	Q (10^4 Btu/hr)
C1	120	235	20000	230
C2	180	240	40000	240
H1	260	160	30000	300
H2	250	130	15000	180

It is assumed that the heat-capacity flow rate, $mC_p = C$, which is the product of the specific heat and the mass flow rate, does not vary with temperature.
Design a HEN that uses the smallest amounts of heating and cooling utilities possible using the HCC method in Excel software using Solver add-on.
For the temperature range in this example, a reasonable assumption is $\Delta T_{min} = 10$ °F.

Solution:

- Note down the total required heat load for the hot fluids (H1+H2) and cold fluids (C1+C2) from Table 5.
- Calculate the heat load for the hot fluids H1 and H2 and plot the cooling curves as shown in Figure 13.

Table 6. Calculated head load for hot fluids H1 and H2.

	Heat load Q (10^4 Btu/hr)	Temp (° F)	mCp ((Btu/(hr.° F))
H1	0	160	30000
	300	260	
H2	0	130	15000
	180	250	

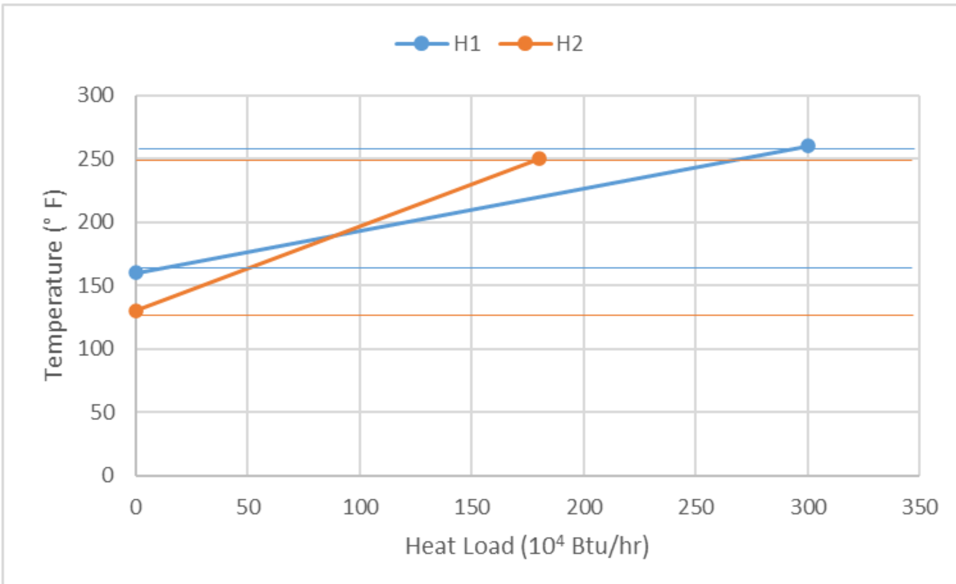

Figure 13. Cooling curves for hot fluids H1 and H2.

- From Figure 13, find the three temperature intervals, calculate the temperature difference of each interval and the heat load in each interval (Qint) using mCpΔT.

Table 7. Interval Heat loads, where Qint is the heat load of each temperature interval for hot fluid.

Interval	Int temp	ΔT	Qint (10^4 Btu/hr)
1	260-250	10	30
2	250-160	90	405
3	160-130	30	45

- From the total required heat load for the hot fluids (H1+H2) of 470 10^4 Btu/hr, subtract the Qint, the value obtained is the cumulative heat load (Qcum). For the final interval the Qcum should be 0.
- Plot the temperature versus Qcum obtained in Table 8 to obtain the heating fluid composite curve.

Table 8. Cumulative heat load for each temperature interval for hot fluid.

Temp (° F)	Qcum (10^4 Btu/hr)
260	480
250	450
160	45
130	0

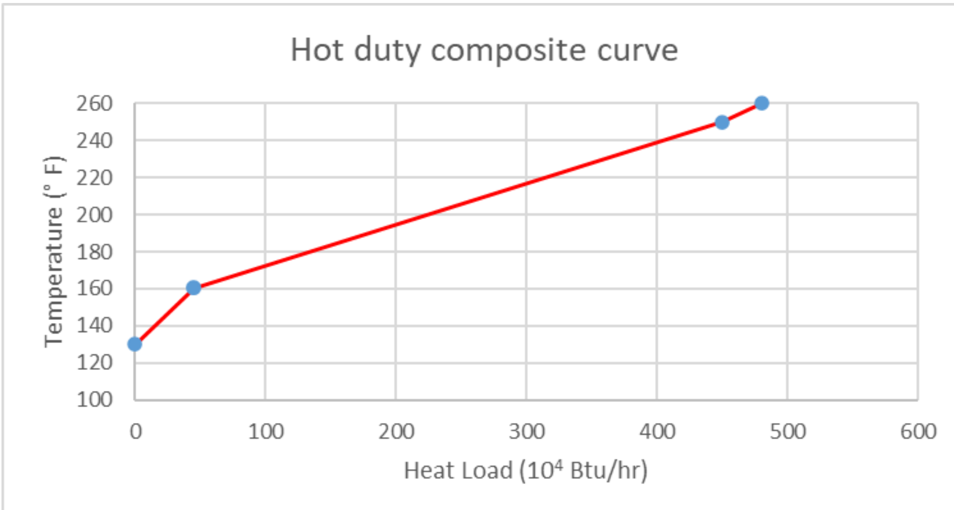

Figure 14. Hot duty composite curve.

- Repeat the same steps as above for the cold fluid.

Table 9. Calculated head load for cold fluids C1 and C2.

	Heat load Q (10^4 Btu/hr)	Temp (° F)	mCp ((Btu/(hr.° F))
C1	0	120	20000
	230	235	
C2	0	180	40000
	240	240	

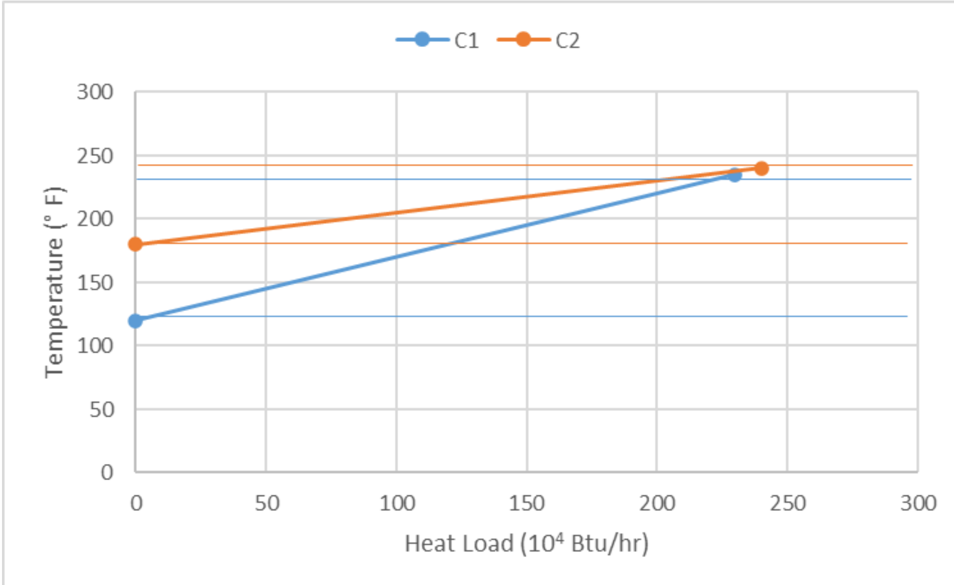

Figure 15. Heating curves for cold fluids C1 and C2.

Table 10. Interval Heat loads, where Qint is the heat load of each temperature interval for cold fluid.

Interval	**Int temp**	**ΔT**	**Qint (10^4 Btu/hr)**
1	240-235	5	20
2	235-180	55	330
3	180-120	60	120

Table 11. Cumulative heat load for each temperature interval for cold fluid.

Temp (° F)	**Qcum (10^4 Btu/hr)**
240	470
235	450
180	120
120	0

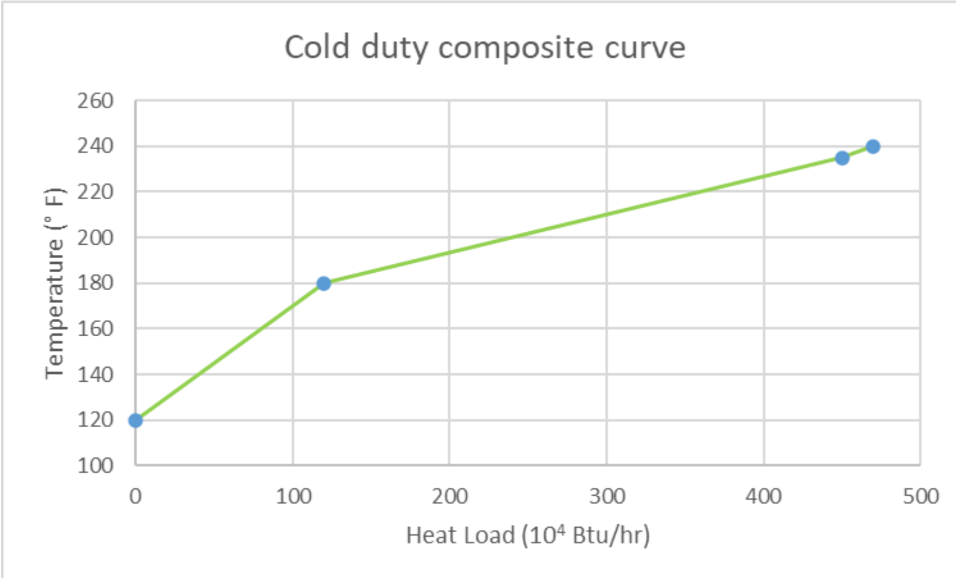

Figure 16. Cold duty composite curve.

- Plot the temperature versus Qcum obtained in Table 8 and Table 11 for the hot fluid and cold fluid on the same graph to obtain the Hohmann/Lochart Composite Curves (HCC).
- The composite curve for the hot fluid and the cold fluid intersects each other 180 °F, where ΔT_{min} = 0.
- If ΔT_{min} is reduced to zero, the cold composite curve is shifted to the left until it touches the hot composite curve, this corresponds to an infinite area for heat exchange.

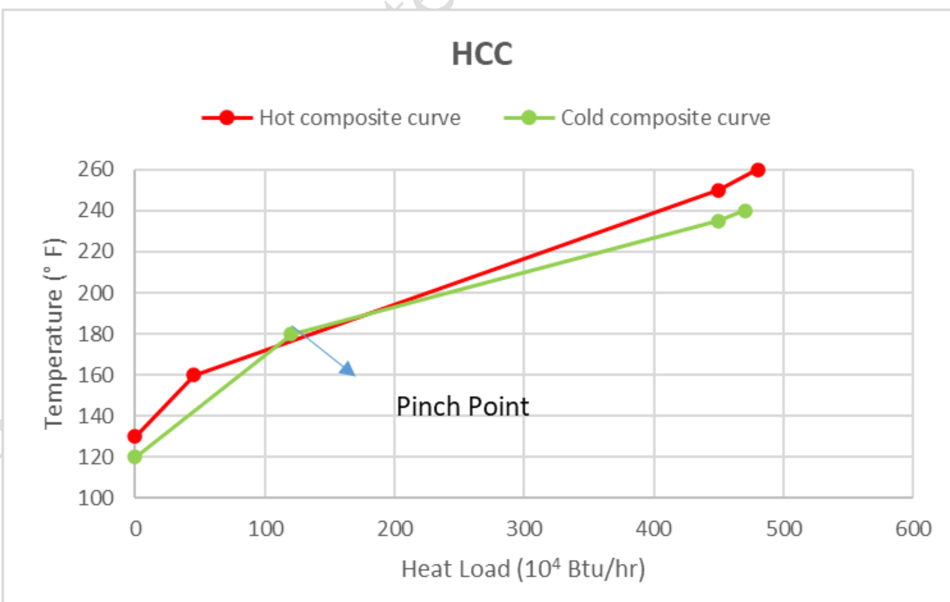

Figure 17. Initial HCC.

- The composite curves should have a closest point of approach of ΔT_{min} = 10°F at the point where stream C2 begins along the cold composite curve; 180°F. These temperatures are the pinch point of cold fluid and hot fluid.
- Use the solver function in Excel (in Data tab) to change the pinch temperature. of cold pinch point.
- Change the value of Qcum for cold pinch interval to maximum value by changing cold pinch temperature (180 °F).
- Keep constraint that cold pinch temp. change cannot exceed cold pinch + ΔT_{min} (190 °F) (as shown in Figure 18).
- Using solver will give new value of Qcum, compare to old original value and find difference.
- Add the difference to all the old Qcum values and plot the shifted cold composite curve using this data.

Table 12. Application of solver to obtain revised Qcum for the HCC (Hohmann/Lochart Composite Curves).

					Qcum	Qint diff	Revised Qint
Interval		**Int temp**	**ΔT**	**Qint (10^4 Btu/hr)**	470		530
1	240	235	5	20	450	60	510
2	235	190	45	270	180		180
3	180	120	60	120	60		60

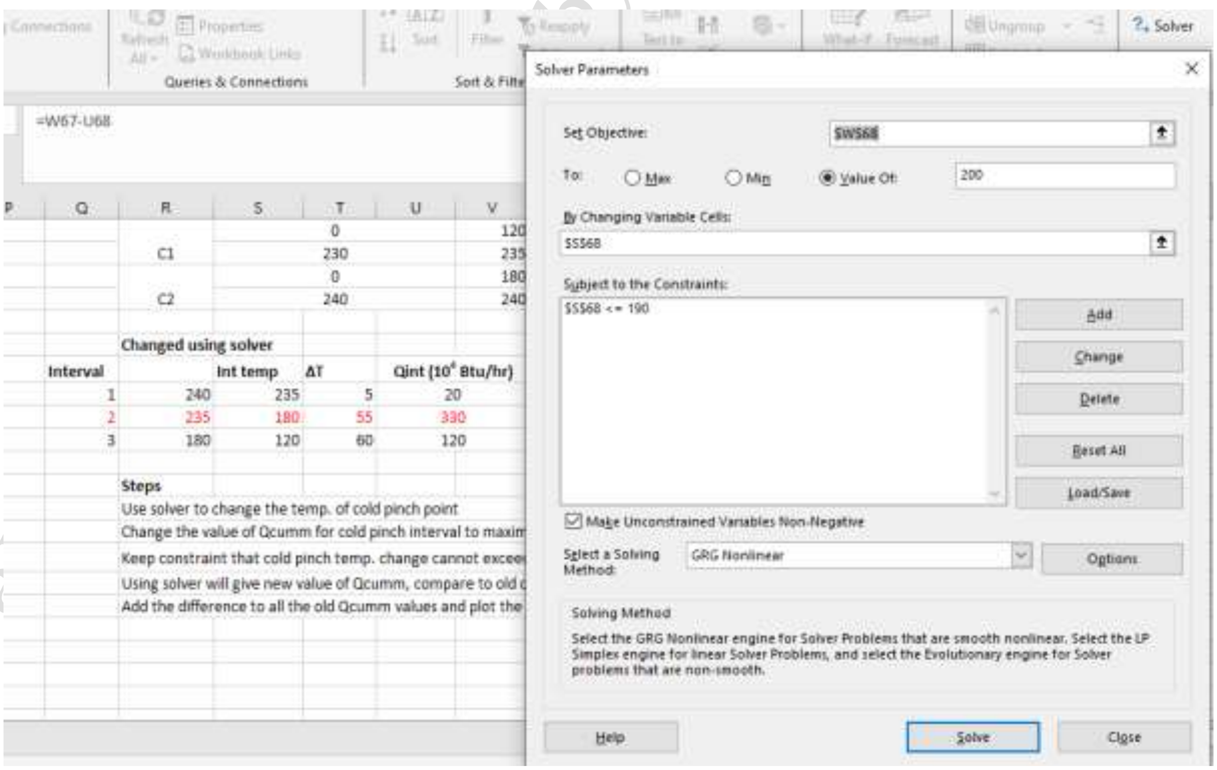

Figure 18. Application of solver to obtain revised Qcum.

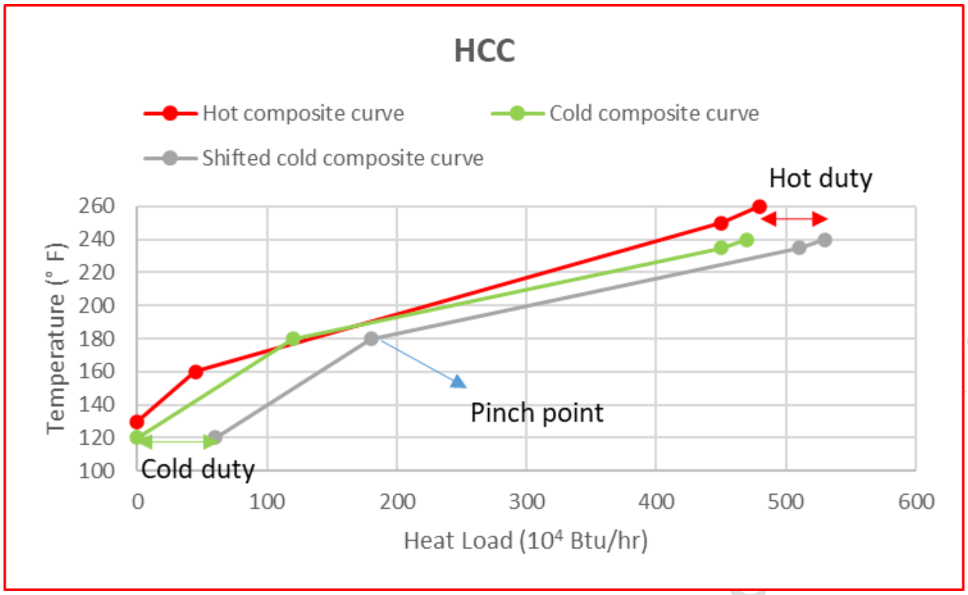

Figure 19. Hohmann/Lochart Composite Curves (HCC) to obtain hot duty Q_{steam}, cold duty Q_{cw} and pinch point.

Thus, using the HCC, an additional 50 × 10^4 Btu/hr must be supplied from a hot utility, such as steam and an additional 60 × 10^4 Btu/hr must be removed using a cold utility, such as cooling water.

Assignment:

Two cold streams, Cl and C2, are to be heated and two hot streams, Hl and H2, are to be cooled without phase change.

Table 13. Conditions and properties of cold streams C1 and C2 and two hot streams H1 and H2.

Stream	T^s (° C)	T^t (° C)	mC_p (kW)	Q (kW)
C1	40	150	3	?
C2	70	130	1	?
H1	170	40	4	?
H2	200	60	6	?

It is assumed that the heat-capacity flow rate, mC_p = C, which is the product of the specific heat and the mass flow rate, does not vary with temperature.

Design a HEN that uses the smallest amounts of heating and cooling utilities possible using the HCC method in Excel software using Solver add-in.

For the temperature range in this example, a reasonable assumption is ΔT_{min} = 10°F.

Results and Discussions:

- In Excel, plot the hot fluid composite curve and the cold fluid composite curve using the steps described above.
- Plot the initial Hohmann/Lochart Composite Curves (HCC).
- Use solver to find the revised Qcum and plot the Hohmann/Lochart Composite Curves (HCC) to obtain hot duty Q_{steam}, cold duty Q_{cw} and pinch point.

- Discuss what would be the effect on hot duty Q_{steam}, cold duty Q_{cw} and pinch point if ΔT_{min} is +20 or -20 °F.
- Perform all the calculations in Excel, submit the Excel file by email.
- Prepare a report as shown in the manual for the key calculations and graphs and submit the hard copy along with manual.

Theroy 2: Grand Composite Curve (GCC):

In the grand composite curve (GCC), the enthalpy residuals are displayed as a function of the interval temperatures. The enthalpy residuals corresponding to the highest and lowest interval temperatures are the minimum heating and cooling utility duties. Furthermore, the process is divided into a portion above the pinch (indicated by a solid circle) in which there is excess heating demand, and a portion below the pinch in which there is excess cooling demand.

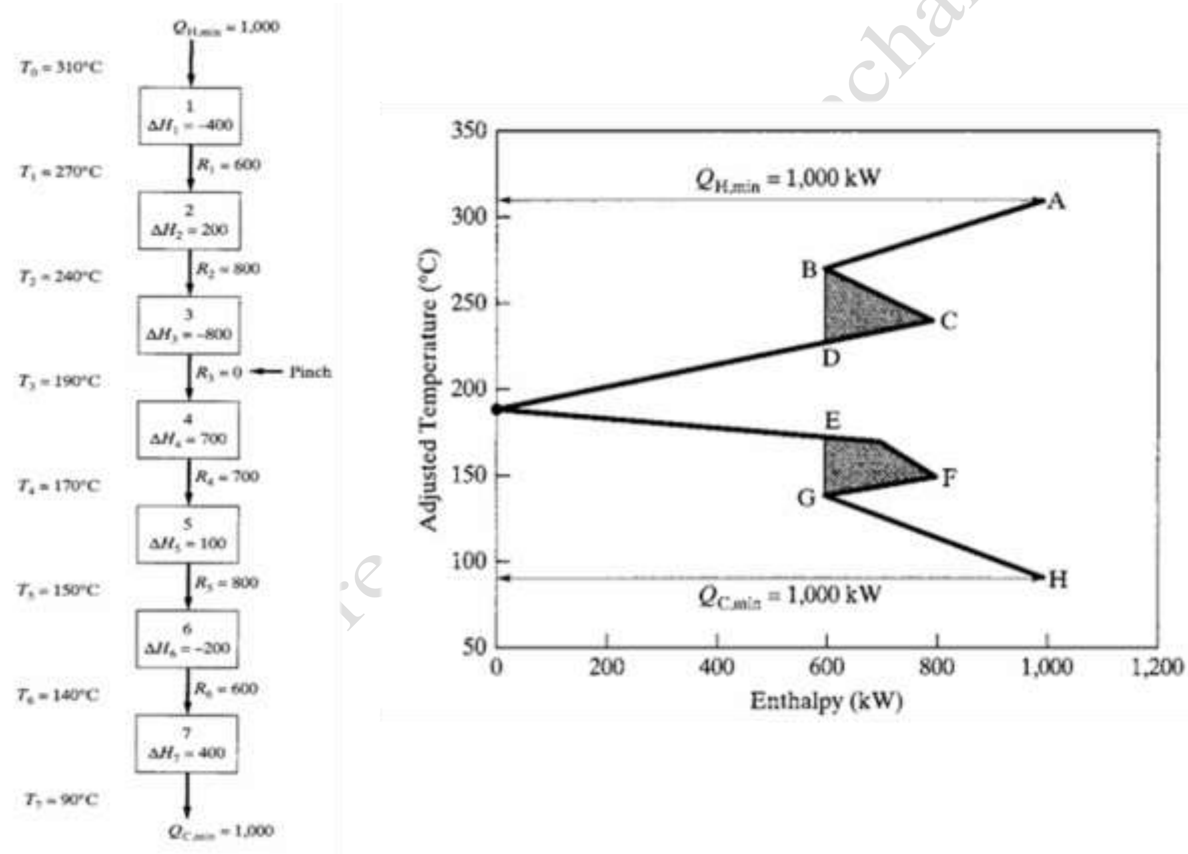

Figure 20. Grand composite curve for temperature intervals, energy balances, and residuals determined in (a).

In addition to this information, Figure 20 provides the following insights:

a. The local rise in the enthalpy residual along the arc BC, which indicates an increase in the residual heat available, can be used to supply the increased demand denoted by the decrease in the enthalpy residual along the arc CD. This is accomplished using internal heat exchange. Similarly, the heat demand along the arc FG can be supplied by internal heat exchange with the arc EF.

b. When 1,000 kW are provided at 330°C, a fired-heater is required that burns fuel. Alternatively, the

GCC replotted in Figure 10.36 shows that up to 600 kW can be provided at 230°C using less expensive high-pressure steam at 400 psi, noting that the alternative utility temperature level and duty are limited by the intercept D. This leaves only 400 kW to be supplied in the fired-heater. Similarly, instead of removing 1,000 kW at 90°C using cooling water, up to 600 kW can be removed at 170°C by heating boiler feed water, leading to further savings in operating costs. As before, the temperature level and duty are limited by intercept E. Evidently, significant savings in operating costs are revealed by the GCC.

The grand composite curve helps to better utilize utility resources when designing HENs.

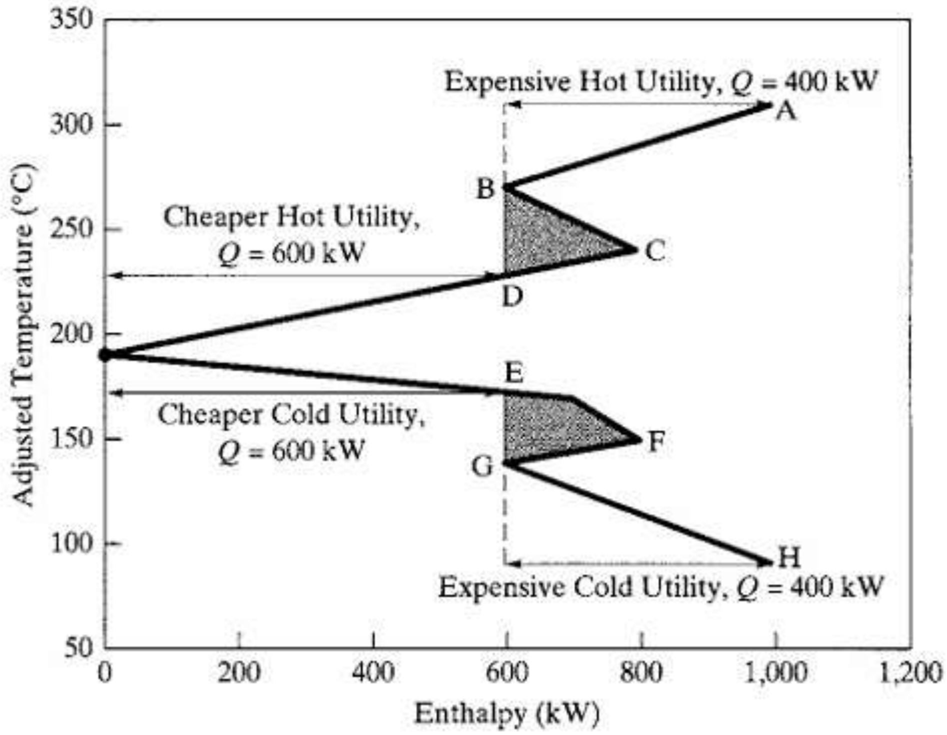

Figure 21. Grand composite curve for the temperature intervals showing possible savings by utilizing HPS and BFW.

Problem Statement: Example 10.17 from Textbook:

Consider the design of a HEN for the four streams below, with ΔT_{min} = 10°C.

Table 14. Conditions and properties of cold streams C1 and C2 and two hot streams H1 and H2.

Stream	T^s (° C) Source temp.	T^t (° C) Target temp.	C (kW/° C)
H1	180	40	20
H2	160	40	40
C1	60	220	30
C2	30	180	22

To reduce operating costs, the design should consider alternative utility sources: high-pressure steam (HPS) and intermediate-pressure steam (IPS), boiler feed water (BFW), and cooling water.

Solution:

The TI method is used to construct the grand composite curve shown in Figure 22, which indicates that MER targets are $Q_{H,min}$ = 2,360 kW and $Q_{C,min}$ = 1,860 kW, with pinch temperatures at 150 and 160°C.

YouTube Link:

https://www.youtube.com/watch?v=6h6rgrDsAq8&list=PL_UhBh-E8IOLmkctKJng1EmQru3RDZyV7&index=3

Solution in Excel:

- First calculate the heat load for the hot fluids (H1 and H2) and cold fluids (C1 and C2).
- Heat load for hot fluids is 7600 kW and heat load fod cold fluids is 8100 kW, there is a difference of 500 kW.

Table 15. Calulation of cold and hot heat load

Stream	T^s (° C) Source temp.	T^t (° C) Target temp.	C (kW/° C))	Q (kW)
H1	180	40	20	2800
H2	160	40	40	4800
C1	60	220	30	4800
C2	30	180	22	3300

- Reduce the temperature of the hot fluids by ΔT_{min} = 10 °C to obtain the adjusted temperatures.
- Draw the cascade of temperature intervals and show the flow of hot and cold streams.

Table 16. Adjusted temperatures of hot and cold fluids.

Stream	T^s (° C) Source temp.	T^t (° C) Target temp.	C (kW/° C))
H1	170	30	20
H2	150	30	40
C1	60	220	30
C2	30	180	22

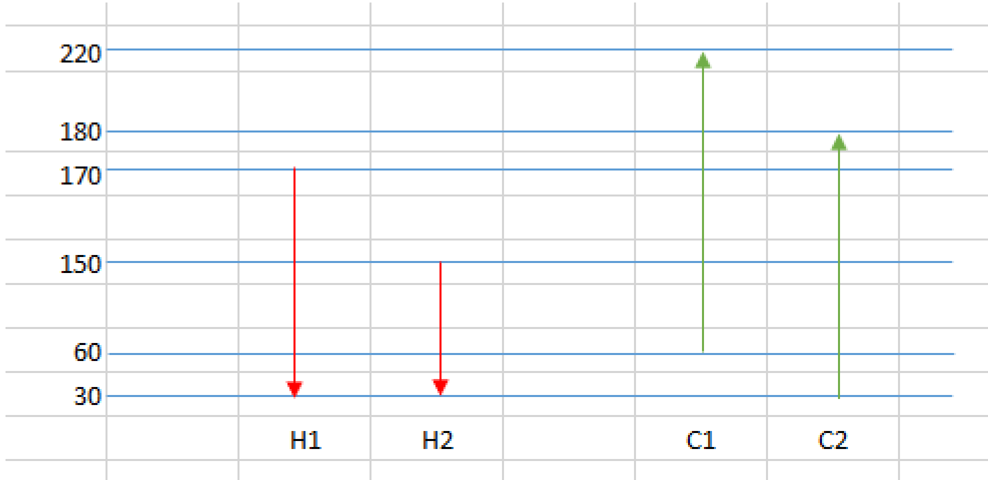

Figure 22. Cascade of temperature intervals with flow of hot and cold stream shown.

- Similar to the temperature interval method, calculate the heat load in the cascade of temperature intervals, energy balances, and residuals. Select the highest negative residual enthalpy (hot duty Qsteam) and add the value to the final pass to obtain the cold pinch point where the residual is 0. The hot pinch point is the cold pinch point added to the ΔTmin. The last residual enthalpy in the cascade is the cold duty (Qcw).

	Qsteam		Residaul		Final Pass			
°C		kW	kW		kW			
220	1				2360			
	-1200		-1200		1160			
180	2							
	-520		-1720		640			
170	3							
	-640		-2360		0	Pinch point	150	°C
150	4					Cold Pinch point	150	°C
	720		-1640		720			
						Hot Pinch point	160	°C
60	5							
	1140		-500		1860			
30								
	Qcw							
			Qsteam	2360	kW			
			Qcw	1860	kW			

Figure 23. Cascade of temperature intervals, energy balances, and residuals.

- For the grand composite curve (GCC), plot the adjusted temperature versus the residual enthalpy in the final pass.

Table 17. Data of adjusted temperature and residual enthalpy for the grand composite curve.

Adjusted Temp. (° C)	Residual enthalpy (kW)
220	2360
180	1160
170	640
150	0
60	720
30	1860

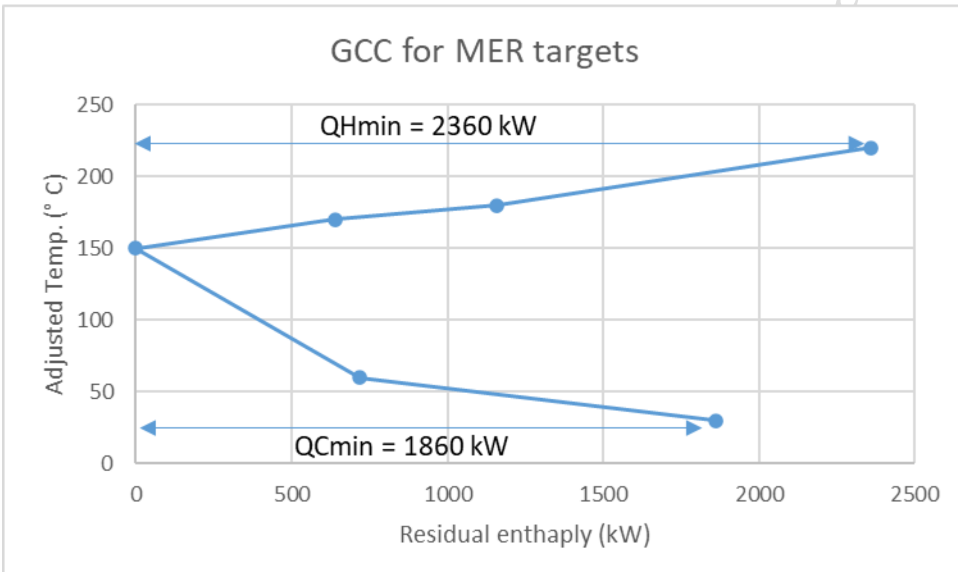

Figure 24. Grand Composite Curve (GCC) with MER targets.

- From the Grand Composite Curve, find the maximum utility that can be used to heat cold and hot streams by internal energy exchange, and obtain the needed external utilities (high-pressure steam (HPS) and intermediate-pressure steam (IPS), boiler feed water (BFW), and cooling water).

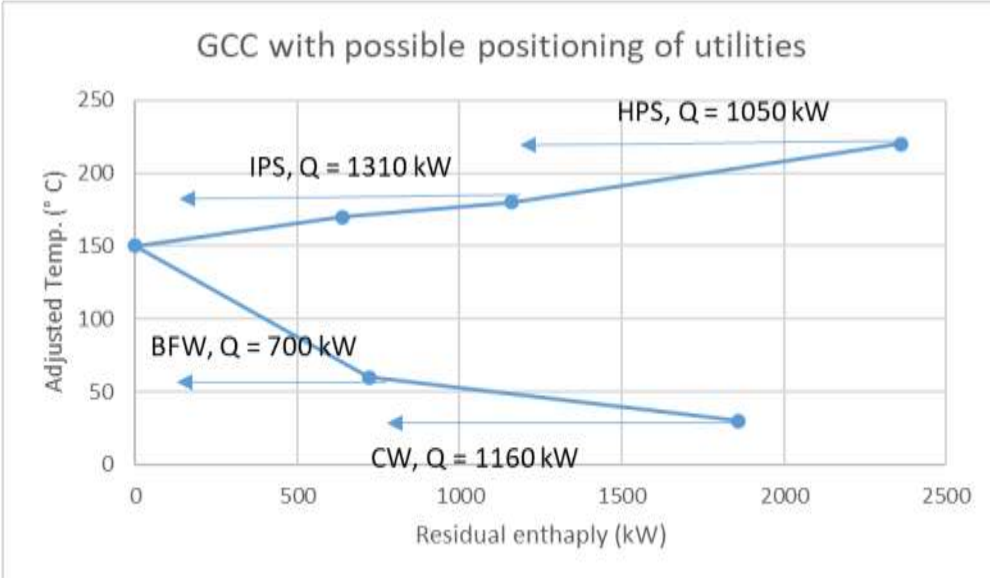

Figure 25. Grand Composite Curve (GCC) with the positioning of utilities.

Assignment:

Consider the design of a HEN for the four streams below, with ΔT_{min} = 10°C.

Table 18. Conditions and properties of cold streams C1 and C2 and two hot streams H1 and H2.

Stream	T^s (° C)	T^t (° C)	mC_p (kW)
C1	40	150	3
C2	70	130	1
H1	170	40	4
H2	200	60	6

To reduce operating costs, the design should consider alternative utility sources: high-pressure steam (HPS) and intermediate-pressure steam (IPS), boiler feed water (BFW), and cooling water.

Result and Discussions:

- Find the heat duty load of hot and cold fluids.
- Using the temperature interval method, calculate the heat load in the cascade of temperature intervals, energy balances, and residuals.
- Plot the GCC with MER using the adjusted temperature and the residual enthalpy in the final pass.
- Find the positioning of alternative utility sources.
- Submit Excel file by email. Submit report in lab manual.

Conclusion:

Experiment No. 5: Minimum utility target and pinch point using LP in MS Excel

Aim: To determine the Minimum utility target and pinch point using Linear Programming method in MS Excel.

Theory:

Linear programming can be used for optimization problems. The three main steps are: model development, optimization (here using solver), and evaluation of solution (analysis).

Data needed for the linear model:

1. Objective function to maximize or minimize
2. Input variables
3. Decision variables
4. Constraints

Problem statement: Example 10.4 from Textbook:

Two cold streams, Cl and C2, are to be heated and two hot streams, Hl and H2, are to be cooled without phase change.

Table 19. Conditions and properties of cold streams C1 and C2 and two hot streams H1 and H2.

Stream	T^s (° F) Source temp.	T^t (° F) Target temp.	mC_p (Btu/(hr.° F))	Q (10^4 Btu/hr)
C1	120	235	20000	230
C2	180	240	40000	240
H1	260	160	30000	300
H2	250	130	15000	180

It is assumed that the heat-capacity flow rate, $mC_p = C$, which is the product of the specific heat and the mass flow rate, does not vary with temperature.

Design a HEN that uses the smallest amounts of heating and cooling utilities possible, such that the closest approach temperature differences never fall below a minimum value. For the temperature range in this example, a reasonable assumption is $\Delta T_{min} = 10$°F. Use the linear programming method.

Solution:

First, the temperature interval method is used to obtain the cascade of temperature intervals, energy balances, and residuals. These values are then used to minimise the heating and cooling duty loads. Qsteam is used as the objective function as when Qsteam is minimum, Qcw is also minimum. Conditions (equality constraints) are energy balances for each of the temperature intervals in the energy cascade formulated in the temperature interval method.

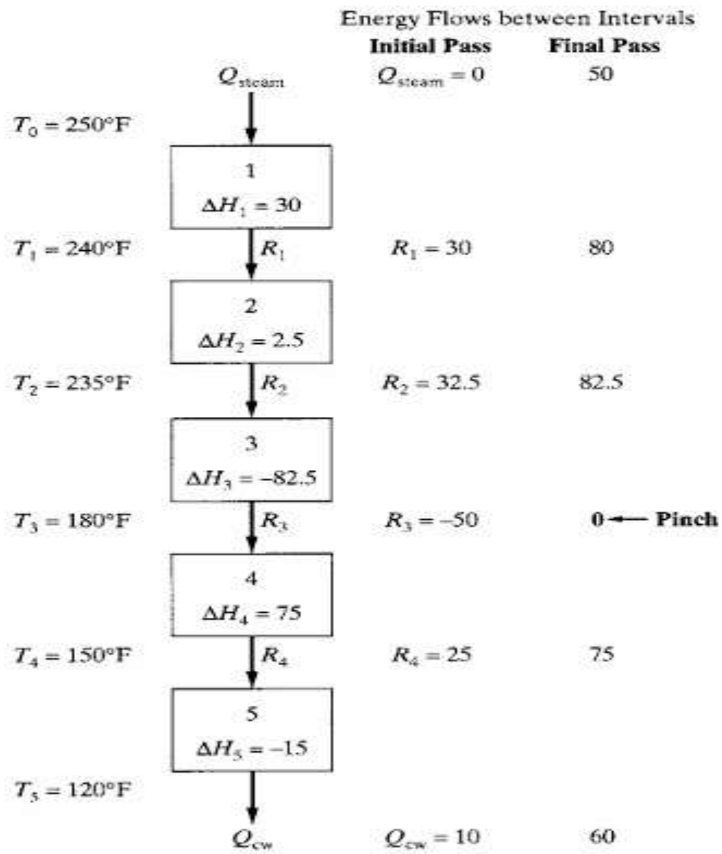

Figure 26. Cascade of temperature intervals, energy balances, and residuals; multiply Q values by 10^4 Btu/hr.

The Linear Programming is formulated:

- Minimize Qsteam

Subject to

- Qsteam – R1 + 30 = 0
- R1 – R2 + 2.5 = 0
- R2 – R3 – 82.5 = 0
- R3 – R4 + 75 = 0
- R4 – Qcw – 15 =0
- Conditions: Qsteam, Qcw, R1, R2, R3, R4 ≥ 0

Solver function in MS Excel will be used for the linear programming method, as per the following steps.

- Add the values of initial variables (any random value) of Qsteam, R1, R2, R3, R4, Qcw.
- Add the equations for the linear programming based on the energy balance from the temperature interval method.

	Variables								
	Qsteam	R1	R2	R3	R4	**Qcw**		**Obj**	
Solution	**100**	200	300	400	500	**600**		-70	
	Equations from enthalpy values								
								LHS	RHS
Constraint 1	1	-1						-100	-30
Constraint 2		1	-1					-100	-2.5
Constraint 3			1	-1				-100	82.5
Constraint 4				1	-1			-100	-75
Constraint 5					1	-1		-100	15

Figure 27. Inputs for linear programming in Excel.

- These equations will become constraints, add the coefficients based on the equations as shown in Figure 27. Use sumproduct function to obtain LHS, write RHS from equations.
- Keep the objective function as equation 1, Qsteam – R1 + 30.
- This objective function has to be minimised using the Solver function.
- Now go to the Data tab and open the Solver function.
- Add the objective function as the cell in which equation 1 is entered.
- Select the minimise option by changing variable cells of Qsteam, R1, R2, R3, R4, Qcw.
- Add the constraints, and tick the non-negativity options for all variables.
- Select solve and the minimum heating duty (Qsteam) and cooling duty (Qcw) will be obtained.

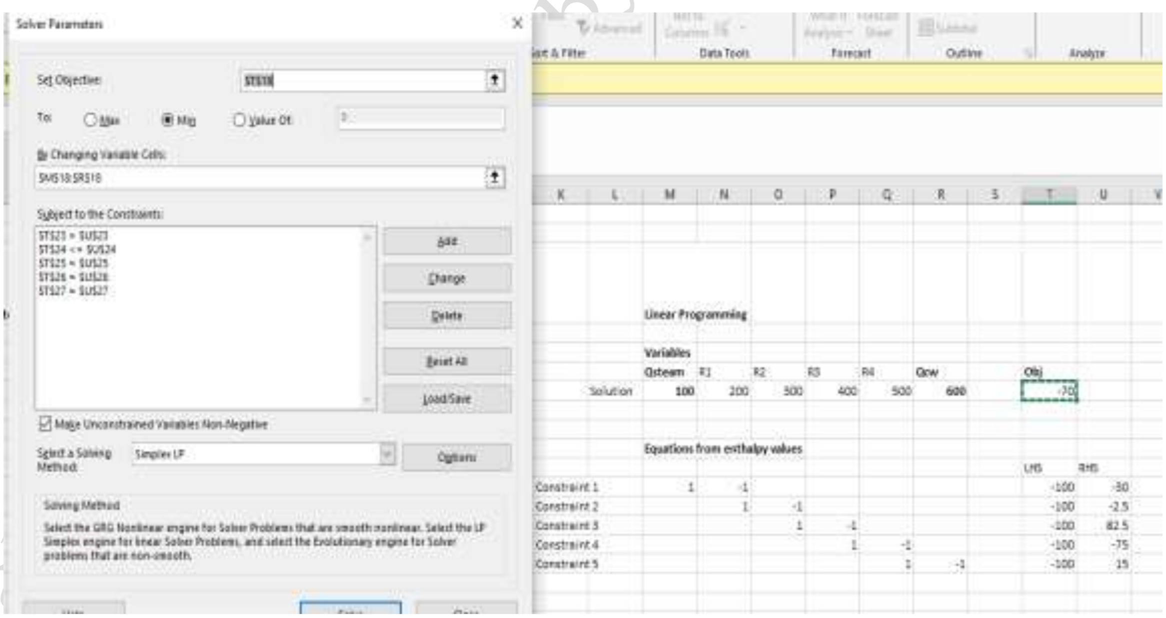

Figure 28. Input for Solver function for linear programming function in Excel.

	Linear Programming							
	Variables							
	Qsteam	R1	R2	R3	R4	**Qcw**		**Obj**
Solution	**50**	**80**	**82.5**	**0**	**75**	**60**		**0**

Figure 29. Results obtained from Solver function.

- The minimum heating and cooling duty are obtained using the linear programming method in Excel as shown in Figure 29.
- These values of minimum heating and cooling duty and enthalpy residuals will match that of the final pass of the temperature interval method.

YouTube Link: https://www.youtube.com/watch?v=ziVlZrTmtI4

Assignment:

Two cold streams, Cl and C2, are to be heated and two hot streams, Hl and H2, are to be cooled without phase change.

Table 20. Conditions and properties of cold streams C1 and C2 and two hot streams H1 and H2.

Stream	T^s (° C)	T^t (° C)	mC_p (kW)	Q (kW)
C1	40	150	3	?
C2	70	130	1	?
H1	170	40	4	?
H2	200	60	6	?

It is assumed that the heat-capacity flow rate, mC_p = C, which is the product of the specific heat and the mass flow rate, does not vary with temperature.

Design a HEN that uses the smallest amounts of heating and cooling utilities possible using the Linear Programming method in Excel software using Solver add-in.
For the temperature range in this example, a reasonable assumption is ΔT_{min} = 10°F.

Results and discussions:

- Determine the cascade of temperature intervals and energy balances using the temperature interval method.
- Add the equations from the energy alances as constrainst in Excel.
- Use the Solver function to minimise the value of Qsteam, and obtain the values of Qsteam, R1, R2, R3, R4, and Qcw.
- Submit Excel file by email. Submit report in lab manual.

Conclusions:

Experiment No. 6: Sequencing of multiple distillation columns

Aim: To sequence multiple distillation columns.

Theory:

Sequencing of Ordinary Distillation for the Separation of Nearly Ideal Fluid Mixtures

Synthesis of ordinary distillation sequences for a multicomponent feed that is to be separated into P final products (pure components and/or multicomponent mixtures). The components in the feed are ordered by volatility, with the first component being the most volatile, consistent with normal boiling point (if mixture forms nearly ideal liquid solutions).

An equation for the number of different sequences of ordinary distillation columns, N_s to produce a number of products, P:

$$\boldsymbol{N_s = \sum_{j=1}^{P-1} N_j N_{P-j} = \frac{[2(P-1)]!}{P!\,(P-1)!}}$$

Table 21. Number of Possible Sequences for Separation by ordinary distillation

Number of Products, P	Number of Separators in the sequence	Number of Difference Sequences, N_s
2	1	1
3	2	2
4	3	5
5	4	14
6	5	42
7	6	132
8	7	429
9	8	1430
10	9	4862

Heuristics for Determining Favourable Sequences:

1. Remove thermally unstable, corrosive, or chemically reactive components early in the sequence.
2. Remove final products one by one as distillates (the direct sequence).
3. Sequence separation points to remove, early in the sequence, those components of greatest molar percentage in the feed.
4. Sequence separation points in the order of decreasing relative volatility so that the most difficult splits are made in the absence of the other components.
5. Sequence separation points to leave last those separations that give the highest-purity products.
6. Sequence separation points that favor near equimolar amounts of distillate and bottoms in each column (cost effective).

Marginal Vapour Rate Method

The marginal vapor rate (MV) method of Modi and Westerberg (1992). Outperforms the other methods and can be applied without the necessity of complete column designs and calculations of costs. For a given split between two key components, Modi and Westerberg consider the difference in costs between the separation in the absence of nonkey components and the sep-aration in the presence of nonkey components, defining this difference as the marginal annu-alized cost (MAC).

They show that a good approximation of MAC is the MV, which is the corresponding difference in molar vapor rate passing up the column. The sequence with the minimum sum of column MVs is selected.

A convenient method for determining the molar vapor rate in an ordinary distillation column separating a nearly ideal system uses the Underwood equations to calculate the minimum reflux ratio, R_{min}.

The design reflux ratio is taken as $R = 1.2*R_{min}$.

By material balance, the molar vapor rate, V, entering the condenser is given by

$$V = D\,(R + 1)$$

where D is the molar distillate rate.

Assuming that the feed to the column is a bubble-point liquid, the molar vapor rate through the column will be nearly constant at this value of V.

Problem Statement: Example 7.1, 7.2, and 7.3:

1. Ordinary distillation is to be used to separate the ordered mixture C_2, $C_3^=$, C_3, $1\text{-}C_4^=$, nC_4 into the three products: C_2; ($C_3^=$, $1\text{-}C_4^=$); (C_3, nC^4). Determine the number of possible sequences.
2. Consider the separation problem shown in Figure 30. Separate isopentane and n-pentane products are also to be obtained with 98% recoveries. Use heuristics to determine a good sequence of ordinary distillation units.

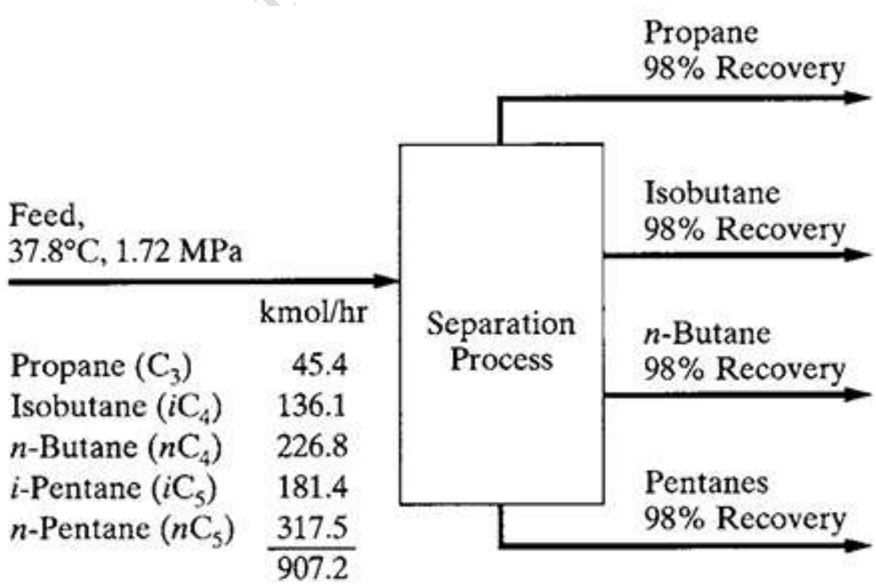

Figure 30. Synthesis problem and separation train.

3. Use the marginal vapor rate (MV) method to determine a sequence for the separation of light hydrocarbons specified in Figure 28, except: (1) remove the propane from the feed, (2) ignore the

given temperature and pressure of the feed, and (3) strive for recoveries of 99.9% of the key components in each column. Data from a process simulation program, with the Soave-Redlich-Kwong equation of state for K-values and enthalpies, to set top and bottom column pressures and estimate the reflux ratio with the Underwood equations.

Table 23. Input data of calculations of marginal vapor rate, MV

Separation	Column Top Pressure (kPa)	Distillate Rate, D (kmol/hr)	Reflux Ratio ($R = 1.2\ R_{min}$)	Vapor Rate, $V = D(R+1)$	Marginal Vapor Rate (kmol/hr)
A/B	680	136.2	10.7	1594	0
A/BC	680	136.2	11.9	1757	163
A/BCD	680	136.2	13.2	1934	340
B/C	490	226.8	2.06	694	0
AB/C	560	362.9	1.55	925	231
B/CD	490	226.8	3.06	921	227
AB/CD	560	362.9	2.11	1129	435
C/D	210	181.5	13.5	2632	0
BC/D	350	408.3	6.39	3017	385
ABC/D	430	544.4	4.96	3245	613

Solution:

1. For the first problem, neither multicomponent product C^2, $C_3^=$, C_3, 1-$C_4^=$, nC_4 contains adjacent components in the ordered list: C_2; ($C_3^=$, 1-$C_4^=$); (C_3, nC_4).

   Therefore, the mixture must be completely separated with subsequent blending to produce the ($C_3^=$, 1-$C_4^=$) and (C_3, nC_4) products.

   $$N_s = \sum_{j=1}^{P-1} N_j N_{P-j} = \frac{[2(P-1)]!}{P!\,(P-1)!}$$

   Thus, from Table 21 with P taken as 5, number of possible sequences $N_s = 14$.

2. For problem 2, 4 products are required after separation, thus the number of possible sequences $N_s = 5$.

   From Table 22, there are wide variations in both relative volatility and molar percentages in the process feed.

   From the list of rules for Heuristics, the choice is Heuristic 4, which dominates over Heuristic 3 and leads to the sequence shown in Figure 31 (b), where the first split is between the pair with the highest relative volatility. This sequence also corresponds to the optimal arrangement.

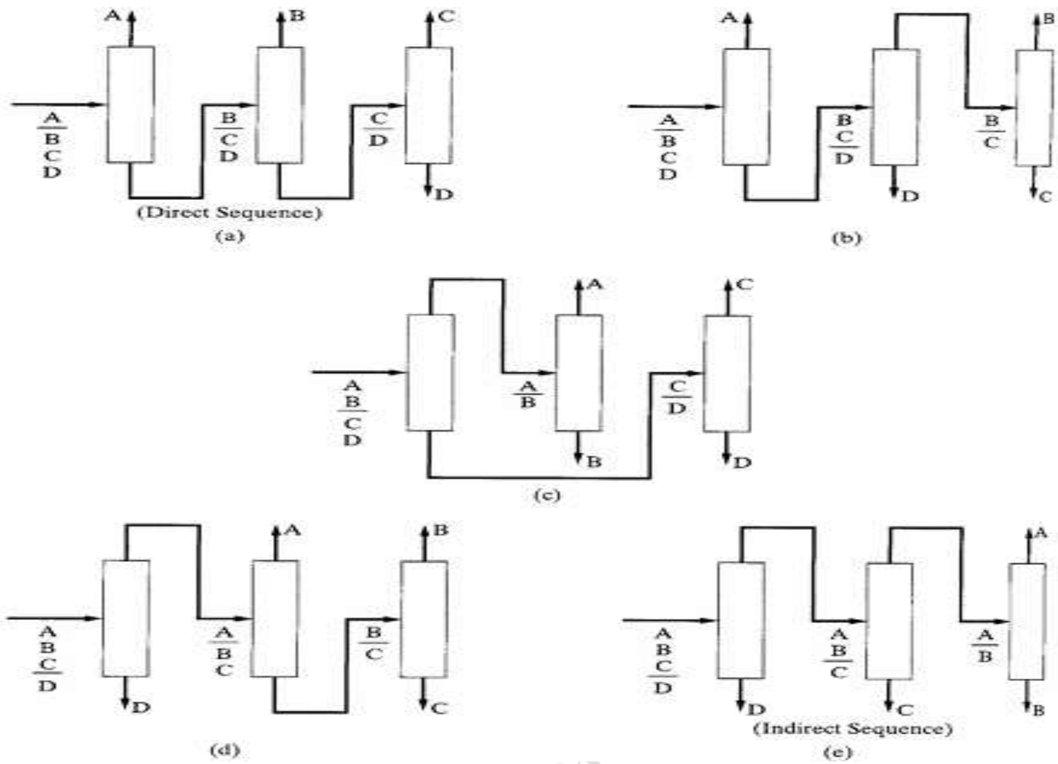

Figure 31. The five possible sequences for a four-component feed.

Table 22. Approximate relative volatilities for all adjacent pairs.

Component pair	**Approximate α at 1 atm**
C3/iC4	3.6
iC4/nC4	1.5
nC4/iC5	2.8
iC5/nC5	1.35

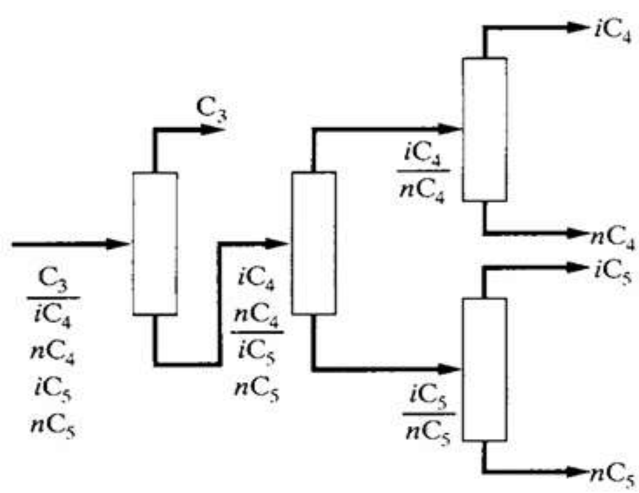

Figure 32. Sequence developed from Heuristic 4.

For the third problem, to produce four nearly pure products from the four-component feed, five sequences of three ordinary distillation columns are possible.

$$N_s = \sum_{j=1}^{P-1} N_j N_{P-j} = \frac{[2(P-1)]!}{P!\,(P-1)!}$$

Let A = isobutane, B = n-butane, C = isopentane, and D = n-pentane. A total of 10 unique separations are possible and are listed in Table 23.

The results of the calculations for the top column pressure, P"/P' in kPa; the molar distillate rate, D, in kmol/hr; and the reflux ratio, R, using the shortcut (Fenske-Underwood-Gilliland or FUG) distillation model of the CHEMCAD process simulation program are listed.

This model applies the Underwood equations to estimate the minimum reflux ratio, as described by Seader and Henley (I998). Column feeds were computed as bubble-point liquids at P_{top} + 35 kPa. Also included in Table are values of the column molar vapor rate, V, in kmol/hr and marginal vapor rate, MV, in kmol/hr.

From Table 23, the sum of the marginal vapor rates are calculated for each of the five possible sequences, and the lowest value is selected as the most optimum.

Table 24 shows that the preferred sequence is the one that performs the two most difficult separations, A/B and C/D, in the absence of non-key components. These two separations are far more difficult than the separation, B/C. The direct sequence is the next best.

Table 24. Marginal Vapor Rates for the five possible sequences.

Sequence in Figure 29	**Marginal Vapor Rate, MV (kmol/hr)**
(a) Direct	567
(b)	725
(c)	435
(d)	776
(e) Indirect	844

Assignment:

1. The effluent from a reactor contains a mixture of various chlorinated derivatives of the hydrocarbon RH_3, together with the hydrocarbon itself and HCI. Based on the following information and the heuristics, devise the best two feasible separation sequences. Explain your reasoning. Note that HCl may be corrosive.

   Table 25. Data of flow rate, relative volatility, and purity desired for the mixture of various chlorinated derivatives of the hydrocarbon RH_3.

Species	Flow rate (lbmol/hr)	α, relative to RCl_3	Purity desired
HCl	52	4.7	80 %
RH_3	58	15.0	85 %
RCl_3	16	1.0	98 %
RH_2Cl	30	1.9	95 %
$RHCl_2$	14	1.2	98 %

2. A mixture of alkanes in the following table is to be separated into relatively pure products. The relative volatilities have been calculated on the basis of the feed composition to the sequence and assuming a pressure of 6 bar_g. Use heurestics and marginal vapor method to determine the best possible separation

sequence (Students can perform the assignment manually or by Excel).

Table 26. Data of flow rate and relative volatility for the mixture of alkanes.

Component	Feed flow rate (kmol/hr)	Normal boiling point (K)	Relative volatility	Adjacent relative volatility
Propane (A)	45.4	231	5.78	
				1.94
i-Butane (B)	136.1	261	2.98	
				1.26
n-Butane (C)	226.8	273	2.36	
				1.95
i-Pentane (D)	181.4	301	1.21	
				1.21
n-Pentane (E)	317.5	309	1.00	

Solution:

- When the number of products is 5, there will be 4 separators in the sequence, and the number of possible sequences are 14 (from Table 21), the possible sequences are shown in Figure 33.

$$\boldsymbol{N_s = \sum_{j=1}^{P-1} N_j N_{P-j} = \frac{[2(P-1)]!}{P!\,(P-1)!}}$$

- According to Heuristics, the best possible sequence is no sequence is the optimal one.
- Using the Marginal Vapor Flow method, calculate the marginal flow penalty.

$$Marginal\ Vapor\ Flow\ MVF = \sum_{non-key\ i} V_i = \sum_{non-key\ i} \left| \frac{\alpha_{ij} f_j}{\alpha_{ij} - \left(\frac{\alpha_{LKj} + \alpha_{HKj}}{2}\right)} \right|$$

- Then for each sequence shown in Figure 33, calculate the MVF and determine the optimal sequence which has the lowest MVF value.

Table 27. Marginal flow penalty values

Key split (LK/HK)	$1/2(\alpha_{LK}+\alpha_{HK})$	VA	VB	VC	VD	VE
A/B	4.38	-	-	264.97	69.24	93.93
B/C	2.67					
C/D	1.785					
D/E	1.105					

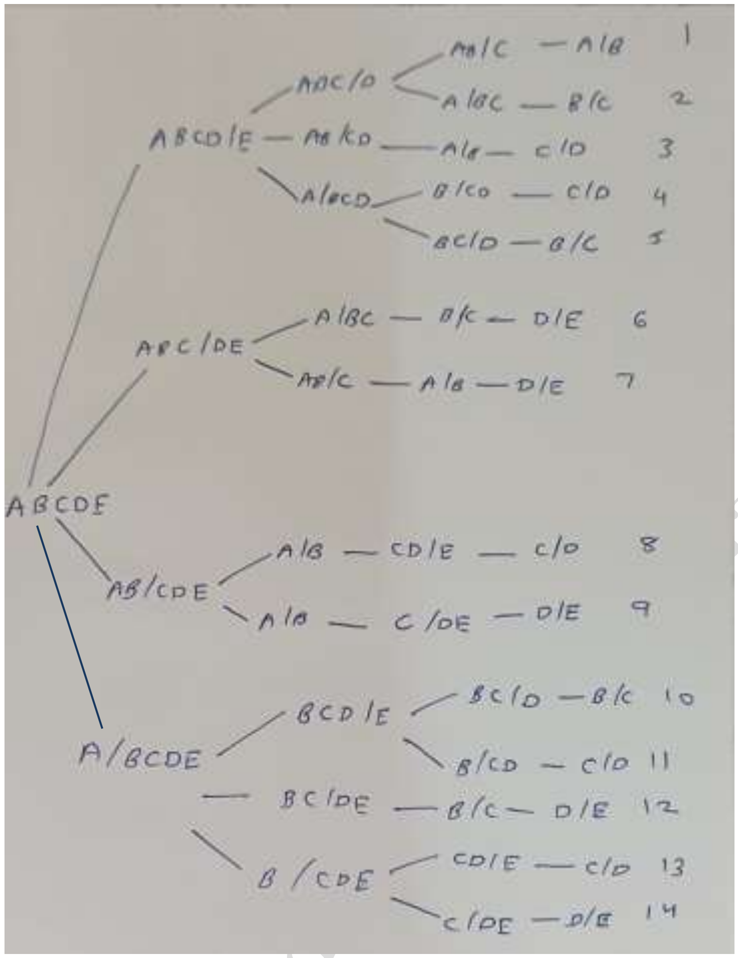

Figure 33. All 14 possible separator sequences for 5 products.

Example: For sequence 1,
ABCD/E – key split D/E, non-key component A, B, C
$MVF_{D/E}$ = 56.13 + 216.31 + 426.49 = 698.93 kmol/hr
ABC/D – key split C/D, non-key components A, B
$MVF_{C/D}$ = 65.69 + 339.40 = 405.08 kmol/hr
AB/C – key split B/C, non-key component A
$MVF_{B/C}$ = 84.38 kmol/hr
Total MVF for sequence 1 = 698.93 + 405.08 + 84.38 = 1188.39 kmol/hr

Table 28. Calculation of MVF for all 14 possible sequences.

Sequence	Combination 1	Combination 2	Combination 3	Total MVF
1	698.93	405.08	84.38	1188.39
2				
3				
4				
5				
6				
7				
8				

9				
10				
11				
12				
13				
14				

YouTube Link: https://media.ed.ac.uk/media/Lecture+8+-+Marginal+Vapour+Flow+Method/1_gq2l1gpb

Results and Discussion:
Make a report and submit it along with manual calculation in lab manual. Submit excel file by email.

Conclusion:

Experiment No. 7: Design and scheduling of batch process

Aim: To design and schedule a batch process for multiple scenarios.

Theory:

Gantt or time event chart is used to visualize the schedule of production, an equipment occupation diagram, showing the periods of time during which each equipment item is utilized. The Batch cycle time is the time interval between successive batches of product being produced. The dead time is the considerable period of time over which the equipment is standing idle which shows poor utilization of equipment.

High utilization of equipment is one of the goals of batch process design, which can be achieved by overlapping batches. This allows the batch cycle time - the time interval between producing successive batches of product, to be decreased considerably. The step with the longest time limits the cycle time. Alternatively, if more than one step is carried out in the same equipment, the cycle time is limited by the longest series of steps in the same equipment. The batch cycle time must be at least as long as the longest step. The rest of the equipment other than the limiting step is then idle for some fraction of the batch cycle.

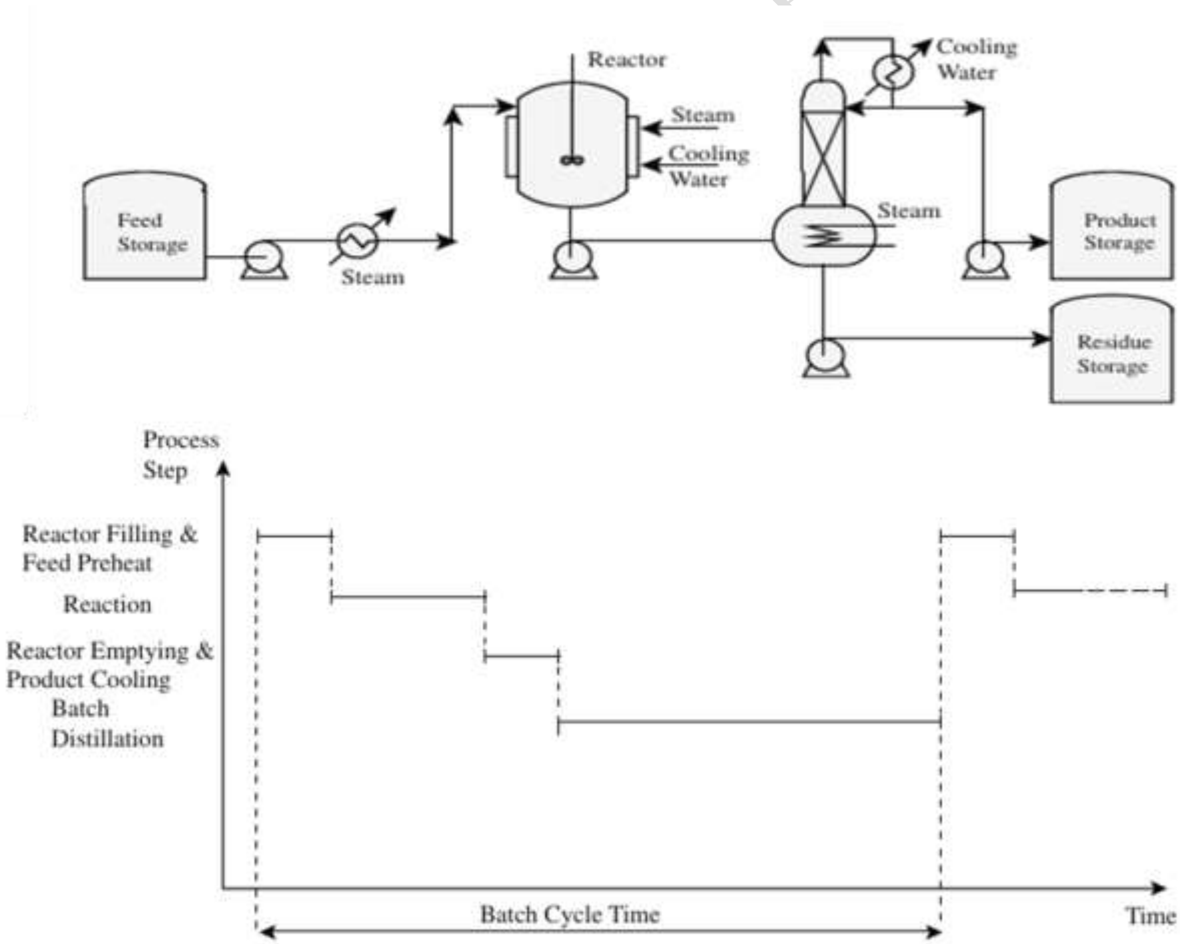

Figure 34. Gantt chart for a simple batch process.

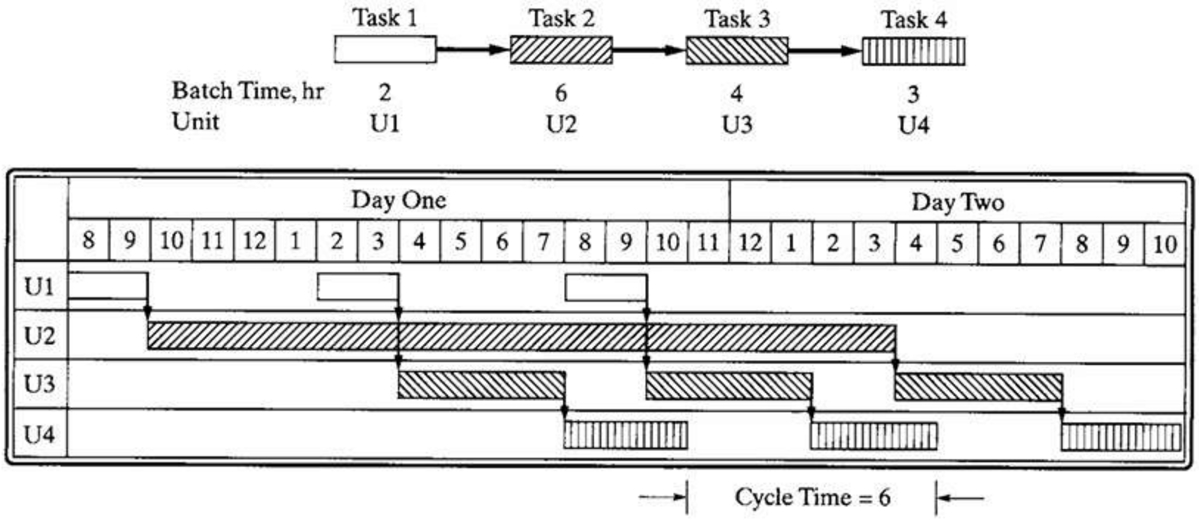

Figure 35. Gantt chart where each distinct task assigned to each unit.

The unit having the longest batch time is the bottleneck unit, as it is always in operation. To increase the efficiency of the schedule and reduce the cycle time, it is common to add one or more units in parallel. When in phase, the batch time for the unit is reduced to $\tau j/nj$, where nj is the number of units in parallel for task j. The parallel units can be sequenced out-of-phase, without altering their batch time.

Intermediate Storage:

Zero-wait (ZW) strategy: contents of each unit are transferred immediately to the next unit, experiencing no delay after its task has been completed

Intermediate storage (IS) strategy: a zero-wait strategy is implemented, with some intermediate storage when necessary

Unlimited intermediate storage (UIS): sufficient to hold the contents of the products from a unit having a lengthy batch time, to be used repeatedly in a unit having half the batch time or less, can greatly reduce cycle time.

Assignment:

Prepare Gantt charts for different scenarios using graph paper:

1. Gantt chart for a simple batch process for four tasks – mixing, heating, reaction, purification.

Table 29. Data for preparation of Gantt chart for a simple batch process

Task	Code	Batch time (hour)	Unit
Task 1		4	U1
Task 2		10	U2
Task 3		3	U3
Task 4		5	U4

2. Gantt chart for a simple batch process for four tasks – mixing, heating, reaction, purification when multiple tasks assigned to same unit.

Table 30. Data for preparation of Gantt chart for a simple batch process

Task	Code	Batch time (hour)	Unit
Task 1		4	U1
Task 2		10	U2
Task 3		3	U3
Task 4		5	U1

3. Gantt chart for above simple batch process for four tasks – mixing, heating, reaction, purification when units can be arranged in parallel.

4. Gantt chart for a multiproduct batch process for 2 products A and B with 2 tasks each.

Table 31. Data for preparation of Gantt chart for a simple batch process

Product	Task	Code	Batch time (hour)	Unit
Product A	Task 1		5	U1
	Task 2		8	U2, U3
Product B	Task 1		6	U1, U2
	Task 2		3	U3

Results and Discussion:

Draw the Gantt charts on graph paper, prepare a report and submit in lab manual.

Conclusion:

Experiment No. 8: Minimum utility target and pinch point and design of heat exchanger network using Software

Aim: Learn how to use software to design heat exchanger networks.

1. Pinch point determination in MATLAB

Reference: Scott Rowe (2025). Pinch (https://www.mathworks.com/matlabcentral/fileexchange/68400-pinch), MATLAB Central File Exchange. Retrieved June 14, 2025.

Website Link: https://in.mathworks.com/matlabcentral/fileexchange/68400-pinch

Steps:

- Open the link and open MATLAB online account.
- Download the files to your directory or add to MATLAB Drive.
- In the command line window, type Pinch and press enter.
- A GUI will open, enter the temperatures (inlet and outlet) of the hot and cold fluids. Specify the value of flowrate and specific heat or add $C = mC_p$.
- Units should be in a consistent format.
- After entering the input data, specify the value of ΔT_{min}, starting from 10 °C.
- Slowly increase the ΔT_{min} and find the threshold and optimum approach temperature, write a report as part of this lab's assignment and submit in lab manual.

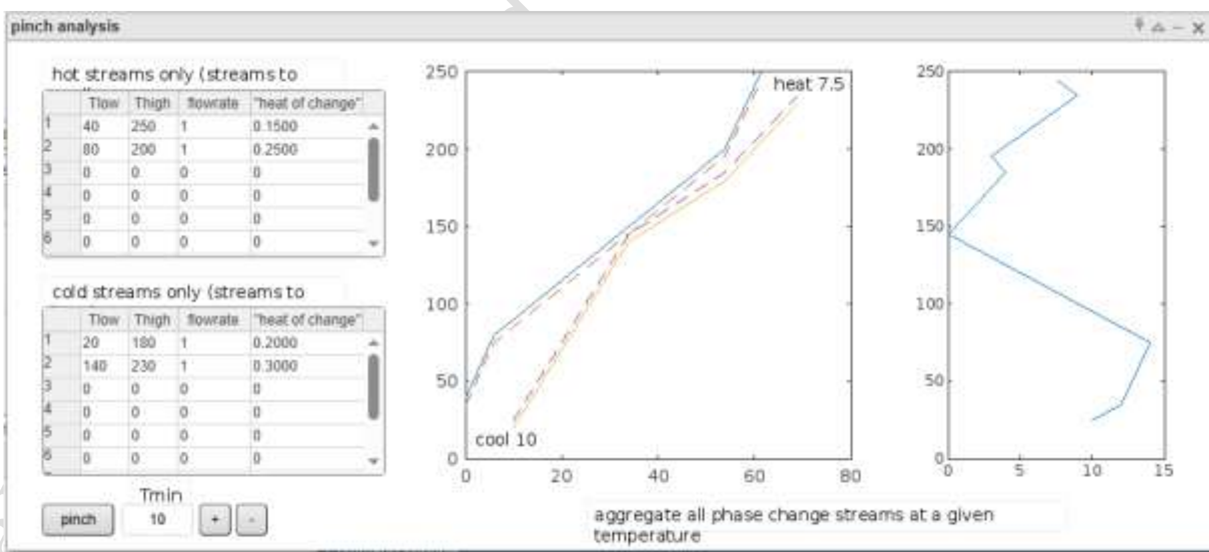

Figure 36. GUI of Pinch determination code in MATLAB (Scott Rowe 2025).

2. Plot Heat Exchanger Network in MATLAB

Reference: David Huber (2025). Plot Heat Exchanger Networks

(https://www.mathworks.com/matlabcentral/fileexchange/130434-plot-heat-exchanger-networks), MATLAB Central File Exchange. Retrieved June 14, 2025.

Code to Plot function for graphical representation of heat exchanger networks as a grid.

Website Link: https://in.mathworks.com/matlabcentral/fileexchange/130434-plot-heat-exchanger-networks

Steps:

- Add the following codes to a script file and observe the different Heat Exchanger Networks (HEN) displayed using the code.
- Play around with the three variables (z, zu, T) and create 3 unique HENs and submit report as part of this lab's assignment in lab manual.

- Variable 1: z - binary matrix representing the heat exchangers

0: no connection

1: connection

dim 1: hot process streams

dim 2: cold process streams

dim 3: steps

- Variable 2: zu - binary vector representing the hot and cold utilities at the stream ends

0: no connection

1: connection

- Variable 3: T - matrix representing the inlet and outlet temperatures of the hot and cold process streams

column 1: inlet temperatures hot streams and cold streams

column 2: outlet temperatures hot streams and cold stream

Code 1:

```
%EX1: HEN with 2 hot, 2 cold process streams and 3 stages
z(:,:,1) = [0,1;0,0]; %hot process streams
z(:,:,2) = [0,0;1,0]; %cold process streams
z(:,:,3) = [1,0;0,1]; %steps
zu = [1;0;0;1]; %hot and cold utilities
T = [400,100;300,250;20,80;15,30]; %c1 - inlet hot and cold temp, c2 - outlet hot and cold temp
[h] = plotHEN(z, zu, T) % call plot function
```

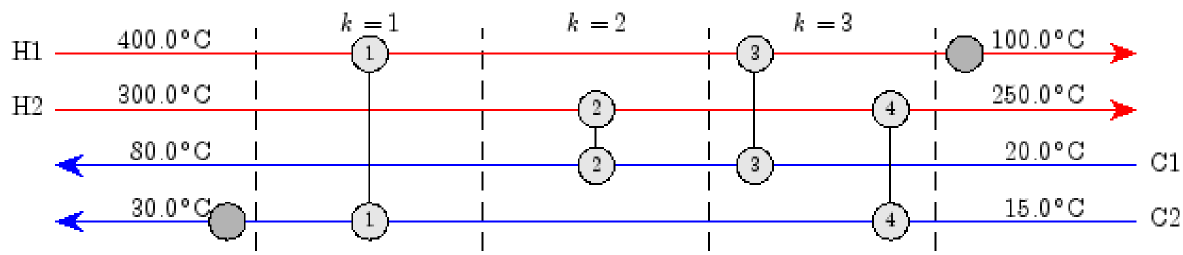

Figure 37. HEN with 2 hot, 2 cold process streams and 3 stages

Code 2:

```
% EX2: HEN with 4 hot, 4 cold process streams and 2 stages
z(:,:,1) = [0,1,0,0;0,0,1,0;0,0,0,0;0,1,0,0];
z(:,:,2) = [0,0,0,0;0,0,0,0;0,0,1,0;1,0,0,1];
zu = [1;0;0;1;1;0;0;1];
T = [400,100;300,250;800,350;600,80;20,60;15,80;35,90;50,300];
[h] = plotHEN(z, zu, T) % call plot function
```

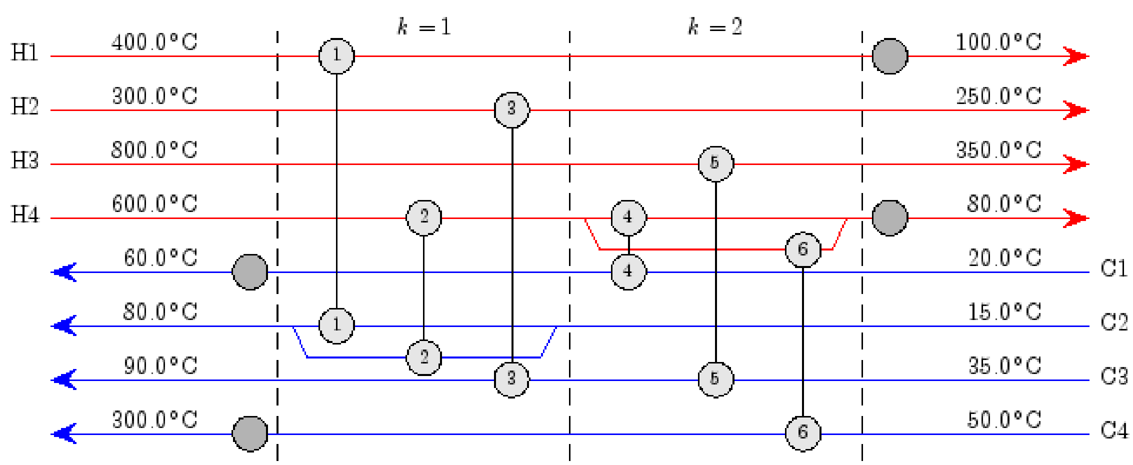

Figure 38. HEN with 4 hot, 4 cold process streams and 2 stages

3. Pinch point determination and design Heat exchanger Network using HINT

Watch the YouTube video for the HINT application on designing Heat Exchanger Networks.

YouTube Link: https://www.youtube.com/watch?v=bXMAJ1DdCbU

Write the steps used in the software to design Heat Exchanger Networks as part of this lab's assignment and submit in lab manual.

Appendix

Basics of MATLAB:

- https://ctms.engin.umich.edu/CTMS/?aux=Basics_Matlab
- https://in.mathworks.com/help/matlab/getting-started-with-matlab.html
- https://www.mccormick.northwestern.edu/documents/students/undergraduate/introduction-to-matlab.pdf